

The Last Word Often Wins: A Format Confound in Chain-of-Thought Corruption Studies

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Abstract

Corruption studies, the primary tool for evaluating chain-of-thought (CoT) faithfulness, identify which chain positions are “computationally important” by measuring accuracy when steps are replaced with errors. We identify a systematic confound: for chains with explicit terminal answer statements, the dominant format in standard benchmarks, corruption studies detect where the answer text appears, not where computation occurs.

A within-dataset format ablation provides the key evidence: on standard GSM8K chains ending with “the answer is X ,” removing only the answer statement, preserving all reasoning, collapses suffix sensitivity $\approx 19\times$ at 3B ($N=300$, $p=0.022$). Conflicting-answer experiments quantify the causal mechanism: at 7B, CC accuracy drops to zero or near-zero (≤ 0.02) across five architecture families; the followed-wrong rate spans 0.63–1.00 at 3B–7B and attenuates at larger scales (0.300 at Phi-4-14B, ≈ 0.01 at 32B). The 7B format-ablation counterpart, a within-stable $9.3\times$ attenuation ($N=76$, $p=7.8\times 10^{-3}$; replicated in Qwen3-8B $N=299$, $p=0.004$), provides converging secondary evidence. The same pattern replicates on MATH (DeepSeek-R1-7B: $10.9\times$ suffix-survival recovery), and on chains without answer suffixes the same protocol identifies the prefix as load-bearing instead ($\Delta=-0.77$, $p<10^{-12}$). Sensitivity follows the answer text, not the computation.

Generation-time probes confirm a dissociation: the answer is not early-determined during generation (early commitment $< 5\%$), yet at consumption time model outputs systematically follow the explicit answer text rather than intermediate reasoning. The format-determination effect persists through 14B ($8.5\times$ ratio, $p=0.001$) even as the direct override attenuates, with both converging toward zero at 32B. We propose a three-prerequisite protocol, question-only control, format characterization, all-position sweep, as a minimal standard for corruption-based faithfulness studies.

1 Introduction

Chain-of-thought (CoT) prompting elicits step-by-step reasoning from large language models [1]. Corruption studies, replacing specific steps with errors and measuring the accuracy drop, are the primary empirical tool for evaluating whether these steps are computationally meaningful [4, 3, 5]. Process reward models [8] and faithfulness evaluations [3] depend on these studies to assign credit to individual reasoning steps.

We identify a systematic confound: for chains with explicit terminal answer statements—the dominant format in benchmarks like GSM8K and MATH—corruption studies detect where the answer text appears, not where computation occurs. Stripping only the answer statement from GSM8K chains while preserving all reasoning collapses suffix sensitivity nearly $20\times$ at 3B (Figure 1). Conflicting-answer experiments confirm the mechanism: models systematically follow wrong terminal answers

over correct intermediate computation, with the effect strongest at 3B–7B and attenuating at larger scales. We distill these findings into a three-prerequisite protocol—question-only control, format characterization, all-position sweep—that should be standard for any corruption-based faithfulness evaluation.

Contributions. (i) We identify and experimentally isolate a previously undocumented format confound in CoT corruption studies. (ii) We provide causal evidence from four complementary designs: within-dataset format ablation, a 2×2 reasoning-by-answer-line factorial, conflicting-answer tests across five architecture families, and answer-placement controls. (iii) We demonstrate that a published-style corruption protocol produces qualitatively different positional conclusions under format control. (iv) Generation-time probes show that answers are not early-determined, bounding the interpretation to answer-text readout dominance at consumption time rather than early answer commitment during generation.

Interpretation. The evidence supports answer-text readout dominance: models can compute through intermediate steps during generation, but at readout their behavioral output preferentially tracks explicit answer text rather than re-deriving the conclusion from reasoning. Whether this reflects a general rationalization pattern or a narrower answer-presentation heuristic shaped by instruction tuning remains an open question (§10).

GSM8K-v1: answer in suffix			GSM8K-stripped: answer removed		
Region	Acc	Δ	Region	Acc	Δ
Baseline	.970	—	Baseline	1.000	—
Prefix	.980	+.010	Prefix	.960	−.040
Middle	.970	.000	Middle	.930	−.070
Suffix	.210	−.760	Suffix	.950	−.050

Qwen 2.5-3B, $N=100$ Qwen 2.5-3B, $N=100$

Figure 1: **Within-dataset format ablation (Qwen 2.5-3B, $N=300$): same model, same examples, same reasoning, only the answer statement removed.** Left: on standard GSM8K-v1 chains where the suffix reads “the answer is X ”, suffix corruption collapses accuracy to 0.210 ($\Delta=-0.760$, $p<10^{-6}$). Right: when only the explicit answer statement is removed from the same chains (GSM8K-stripped-v1), suffix sensitivity collapses $\approx 19\times$ to $\Delta=-0.040$ ($p=0.022$, sign test, $N=300$), identifying answer placement as the mechanism. The only variable is answer placement in the suffix. The cross-dataset version of this reversal (Hard-v3 vs. GSM8K-v1) is shown in Table 3.

2 Related Work

Chain-of-thought faithfulness and causal probes. Turpin et al. [3] showed that CoT can be unfaithful when models are biased by few-shot demonstrations: the generated chain does not always match the factors actually driving the prediction. Ye and Durrett [6] showed that post-hoc explanations from NLP models are unreliable indicators of underlying computation. Lanham et al. [4] showed that truncating or corrupting CoT steps can hurt accuracy, suggesting at least some chain content plays a causal role. Pfau et al. [5] explored hidden computation in transformer language models, showing that filler tokens can serve as implicit computation steps. Baker et al. [15] recently proposed monitoring reasoning faithfulness by comparing the chain-of-thought to a model’s internal activations, providing a complementary perspective to behavioral corruption studies. None of these works control for the confound we identify: that positional corruption sensitivity may reflect answer placement in the chain format rather than computational structure. To our knowledge, no prior work has demonstrated that the same corruption protocol yields opposite positional conclusions on chains with different answer-placement formats, nor has any work proposed a format ablation to isolate the mechanism.

CoT prompting and emergent reasoning. Wei et al. [1] introduced CoT prompting; Wang et al. [2] extended it with self-consistency decoding; Kojima et al. [10] demonstrated zero-shot CoT. Madaan and Yazdanbakhsh [11] decomposed chain contributions (symbols, patterns, text) but did not characterize which positions carry the explicit answer. These works establish that chain content correlates with accuracy but leave causal attribution open.

Theoretical capacity. Merrill and Sabharwal [12] proved that CoT extends transformer capacity beyond TC^0 , but this is a bound on *existence*, not on whether empirical chains use that capacity at the probed positions. Saparov and He [13] showed systematic errors in multi-step inference, suggesting the chain-computation relationship is complex.

Process supervision. Lightman et al. [8] and Uesato et al. [9] assign step-level credit via process reward models, implicitly assuming steps are causally meaningful at their textual positions. Goyal et al. [14] showed that learnable pause tokens provide computation without interpretable steps. Our finding that causal sensitivity tracks answer placement suggests process reward models may disproportionately reward answer expression over reasoning.

Sycophancy and instruction following. A related but distinct phenomenon is sycophancy, where models adapt outputs to match perceived user preferences regardless of correctness [16, 17]. Sharma et al. [17] showed that RLHF training amplifies this tendency, causing models to endorse user-expressed views even when incorrect. Unlike sycophancy studies, which investigate social-desirability bias in open-ended dialogue, our work isolates answer-text *placement* as a methodological confound in chain-of-thought corruption experiments, showing that sensitivity to corrupted reasoning reflects consumption of explicit answer tokens embedded in the chain format, rather than deference to an experimenter’s social preference.

Gap. No prior work has (i) demonstrated that chain format determines positional sensitivity in corruption studies, (ii) proposed format-awareness controls for corruption-based causal analysis, (iii) shown that a format ablation can shift the corruption sensitivity pattern, isolating answer placement as the dominant causal factor, or (iv) provided a direct conflicting-answer causal test showing that models prioritize explicit answer text over their own correct computation at *consumption time*.

3 Problem Formulation

3.1 Answer-Text Readout Dominance

We define the central question this paper investigates. Let q denote a question and $c = (s_1, \dots, s_K)$ a chain of K reasoning steps. A model f maps $(q, c) \rightarrow \hat{a}$ to produce a final answer.

Hypothesis 1 (Answer-Text Readout Dominance). At *consumption time*, when a model reads a completed chain, the model’s final answer $\hat{a}$ is primarily determined by *explicit answer text* present in the chain (e.g., a terminal “the answer is X ” statement), rather than by the intermediate computation encoded in steps $s_1, \dots, s_{K-1}$. This describes a behavioral regularity, the model’s output systematically tracks the explicit answer, without requiring a specific cognitive mechanism (e.g., instruction-following, format-completion heuristics, or recency-weighted readout are all compatible with this regularity). Letting a_{exp} denote the explicit answer text in c :

$$\hat{a} \approx f_{\text{readout}}(q, a_{\text{exp}}),$$

with intermediate reasoning steps playing a reduced causal role at readout, even when they encode the correct answer through genuine step-by-step computation.

If answer-text readout dominance holds, a key behavioral prediction follows: models will be sensitive to corruptions wherever the chain *expresses* the answer text, not wherever intermediate computation *occurs*. This is consistent with a model that genuinely computes through intermediate steps during generation, yet at readout preferentially tracks the explicit answer signal rather than re-deriving the conclusion from embedded reasoning. Our experiments test this prediction directly.

3.2 Notation

Accuracy on a task slice $\mathcal{D} = \{(q_i, c_i, a_i^*)\}_{i=1}^N$ is

$$\text{acc}(f, \mathcal{D}) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[f(q_i, c_i) = a_i^*].$$

Corruption operator. For a position subset $P \subseteq [K]$, define $c^P = \text{corrupt}(c, P)$ as the chain with steps at positions P replaced by semantically incorrect but syntactically valid alternatives (e.g., wrong arithmetic, incorrect logical conclusions). We study three canonical position subsets:

- *Prefix* (P_{pre}): steps in the first third of the chain.
- *Middle* (P_{mid}): steps in the middle third.
- *Suffix* (P_{suf}): steps in the final third.

Question-only condition. Let $f_\emptyset(q)$ denote the model’s answer given *only* the question, with no chain provided. Accuracy under the question-only condition is

$$\text{acc}_\emptyset(\mathcal{D}) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[f_\emptyset(q_i) = a_i^*].$$

3.3 The Question-Solvability Confound

Definition 1 (Question-solvability confound). A corruption study on slice $\mathcal{D}$ suffers the *question-solvability confound* when $\text{acc}(f, \mathcal{D}) - \text{acc}_\emptyset(\mathcal{D}) \leq \epsilon$ for a small threshold $\epsilon \geq 0$.

When the confound holds, observed robustness of $\text{acc}(f(\cdot, c^P), \mathcal{D})$ to corruption cannot be attributed to chain causal structure: the model answers primarily from the question, not the chain.

3.4 Control Requirement and Interpretability

Definition 2 (Control requirement). A corruption study produces interpretable positional evidence only if

$$\text{acc}(f, \mathcal{D}) - \text{acc}_\emptyset(\mathcal{D}) > \epsilon_{\min},$$

where $\epsilon_{\min} > 0$ is a meaningful minimum gap established by a statistical test rejecting $H_0: \text{acc}(f) = \text{acc}_\emptyset$.

Definition 3 (Causally load-bearing position). Under the control requirement, position P is *causally load-bearing* for $\mathcal{D}$ if

$$\text{acc}(f, \mathcal{D}) - \text{acc}(f(\cdot, c^P), \mathcal{D}) > 0$$

with sufficient statistical confidence on a matched paired test.

This formalization requires *two* sequential tests: (i) the chain-gap test (control requirement), then (ii) the positional-drop test (causal load-bearing test). A study reporting only (ii) without (i) cannot support positional causal claims.

3.5 The Answer-Placement Prediction

A key testable prediction of Hypothesis 1 concerns how models behave in corruption studies. If a model is answer-text-prioritizing at readout, it will be sensitive to corruptions at positions where the answer is *expressed*, regardless of where computation logically occurs. This yields:

Hypothesis 2 (Answer-placement prediction). The set of positions identified as “causally load-bearing” by a corruption study is determined by where the chain format explicitly encodes the final answer, not by where genuine intermediate computation occurs.

This hypothesis generates two testable predictions: (a) chains that place the answer in the prefix should show prefix sensitivity; chains that place the answer in the suffix should show suffix sensitivity; and (b) modifying the format (e.g., removing explicit answer statements) should shift the sensitivity pattern.

A third, stronger prediction follows directly from Hypothesis 1:

Hypothesis 3 (Direct conflict prediction). When correct reasoning steps conclude with a conflicting wrong explicit answer, an *answer-text-prioritizing* model will follow the wrong explicit answer (tracking the stated answer text), while a *computation-prioritizing* model will follow its own computation and produce the correct answer (ignoring the stated answer text).

Table 1: Task slices used in this study. “Answer in suffix” indicates whether the chain’s final steps explicitly restate the answer.

Slice	N	Answer in suffix?	Purpose
Hard-v3 (synthetic)	60	No	Prefix-sensitivity test
GSM8K-v1 (benchmark)	100	Yes	Suffix-sensitivity test
GSM8K-stripped-v1	300	Removed	Format ablation
GSM8K-conflict-v1	500	Conflicting	Direct causal test
Commonsense-v1	150	Conflicting	Cross-domain generalization
MATH-v1 (benchmark)	100	Yes ($\backslash$ boxed)	Cross-domain format ablation
MATH-stripped-v1	100	Removed	MATH format ablation

This third prediction directly opposes the two accounts: it cannot be explained by format sensitivity or corruption artifacts, since the steps themselves are intact and correct. Section 7.2 tests this prediction using a three-condition causal experiment.

4 Experimental Setup

4.1 Task Slices

We evaluate four task slices that vary in chain format, difficulty, and scale (Table 1), plus a cross-domain commonsense set.

Primary versus exploratory analysis. The three core pre-specified tests are H1 (answer-line ablation collapses suffix sensitivity), H2 (explicit answer text dominates behavioral output via conflicting-answer protocol), and H3 (positional sensitivity tracks answer placement bidirectionally). Results on additional models, datasets, and subgroups are clearly labeled as replication, sensitivity, or exploratory in the corresponding sections. All primary p -values are pre-specified one-per-endpoint; where multiple comparisons appear within a section, we report Bonferroni-aware interpretations alongside nominal values and label $0.05 < p < 0.10$ as directional.

Hard-v3 (synthetic, $N=60$). A generated 60-example slice designed to suppress question-only shortcuts.¹ Early chain steps explicitly extract relevant quantities into named symbolic variables; middle steps operate on those variables; suffix steps do *not* directly restate the final answer. The answer must be derived from symbolic anchors established in the prefix. The slice spans multiple problem domains with varied distractors. Corrupted steps were generated by GPT-4o with prompt-injection of wrong intermediate values (arithmetic errors), then manually verified to ensure grammatical coherence and absence of correct-answer leakage. The slice was frozen before model evaluation and has not been modified post-hoc. Hard-v3 is used as *corroborating evidence* for the format-determination claim; the primary evidence is the within-GSM8K format ablation (§5) on naturally occurring benchmark chains.

GSM8K-v1 (benchmark, $N=100$). A deterministically filtered 100-example subset of the GSM8K test set [7]. We retain examples with integer-normalized final answers and parsed gold rationales of at least four steps. Gold rationales are segmented into prefix, middle, and suffix thirds. Critically, GSM8K gold chains typically *end* with an explicit answer statement (“The answer is X ”), making the suffix structurally distinct from the synthetic slice.

GSM8K-stripped-v1 (format ablation, $N=300$). GSM8K-v1 examples ($N=300$) with explicit answer statements (e.g., “The answer is 24”) removed from chain suffixes. This creates a matched pair: identical questions, identical intermediate reasoning, but the suffix no longer carries the answer verbatim. If positional sensitivity shifts away from the suffix on this slice, the format-determination hypothesis (Hypothesis 2) is confirmed. For Qwen 2.5-3B, we use $N=300$ for the matched format-ablation comparison (Table 5) and $N=100$ for the cross-format comparison (Table 3).

¹ $N=60$ is the question-only-controlled subset: examples where the question-only accuracy across three probing runs is $\leq 50\%$. See §6.3 for the full QO-control procedure.

Commonsense-v1 (cross-domain, $N=150$). A set of 150 commonsense reasoning examples spanning five domains (social, temporal, counterfactual, spatial, and causal reasoning). Unlike the arithmetic slices, answers are text-based (e.g., “grateful,” “the fire grows”), testing whether the conflicting-answer effect generalizes beyond numeric extraction.

4.2 Models

We evaluate nine open-weight instruction-tuned models from four architecture families:

- *Qwen 2.5-3B-Instruct* (Qwen/Qwen2.5-3B-Instruct), 3.1B parameters
- *Phi-3-mini-4k-instruct* (microsoft/Phi-3-mini-4k-instruct), 3.8B parameters
- *Qwen 2.5-7B-Instruct* (Qwen/Qwen2.5-7B-Instruct), 7.6B parameters (GSM8K-v1 scale evaluation; Appendix A.2)
- *DeepSeek-R1-Distill-Qwen-7B* (deepseek-ai/DeepSeek-R1-Distill-Qwen-7B), 7B parameters; Qwen-2.5-Math-7B fine-tuned on DeepSeek-R1’s reasoning traces (distillation hypothesis probe; Section A.1)
- *Mistral-7B-Instruct-v0.3* (mistralai/Mistral-7B-Instruct-v0.3), 7.2B parameters; third architecture family (Section A.1)
- *Qwen 2.5-14B-Instruct* (Qwen/Qwen2.5-14B-Instruct), 14.7B parameters (scale evaluation; Section A.1)
- *Phi-4* (microsoft/phi-4), 14B parameters; cross-family 14B scale replication (Section A.1)
- *Qwen 2.5-32B-Instruct* (Qwen/Qwen2.5-32B-Instruct), 32.8B parameters (32B scale point; Section A.1)
- *Qwen3-8B* (Qwen/Qwen3-8B), 8B parameters; next-generation Qwen model.
- *Qwen3-14B* (Qwen/Qwen3-14B), 14B parameters; within-Qwen3-generation scale gradient (Section A.1).

Phi-3-mini, Mistral, and Phi-4 provide cross-family replication: the override effect holds across Qwen, Phi, and Mistral architectures, ruling out single-family artifacts. DeepSeek-R1-Distill-7B extends the probe to the training-objective axis: it is a Qwen-architecture model fine-tuned on DeepSeek-R1’s 671B reasoning traces, testing whether reasoning-oriented distillation might transfer resistance to answer-text override (we do not observe this in the current single-pair comparison). The 7B, 14B, and 32B evaluations establish a monotonically decreasing scale-dependent gradient: CC accuracy drops to zero or near-zero (≤ 0.02) at 7B across five architecture families (FW strict = 0.30; corrected = 1.00), dramatically attenuated at 14B (Phi-4: FW = 0.300), and near-zero at 32B (Qwen-32B: FW = 0.010).

4.3 Protocol

Each model is evaluated in five conditions per slice: chain-enabled baseline, prefix-corrupted, middle-corrupted, suffix-corrupted, and question-only. Semantic corruption replaces steps with syntactically valid but arithmetically or logically incorrect alternatives: arithmetic operators are swapped (e.g. $+$ $\rightarrow$ $-$, $\times$ $\rightarrow$ $\div$), numbers are incremented by one, and common math phrases are exchanged (“plus” $\rightarrow$ “minus”). The corrupted step retains its original length and syntactic structure, changing only the mathematical content. Corruption fraction is 100% of eligible steps within the target region to maximize causal signal at these sample sizes. Answer extraction uses a structured step that prefers a clean leading integer line and evaluates simple arithmetic expressions when needed.

Statistical tests use exact paired sign tests per-example and bootstrap 95% confidence intervals (2000 resamples) for accuracy differences. Our inferential hierarchy is pre-specified and fixed across all models: (H1) conflicting-chain accuracy collapse (CC accuracy under conflicting suffixes), (H2) follow-wrong rate (FW), and (H3) position-specific accuracy deltas from corruption. H1 is the primary endpoint because it is extraction-invariant in the sign-negation setting (CC remains 0.00 under both strict and corrected extraction; Appendix C). H2 is the primary effect-size endpoint: FW is first reported under strict extraction, and magnitude-corrected FW is reported as a sensitivity check when sign-negation artifacts are present (§7.2). H3 (position deltas, baseline differences, and cross-format contrasts) is diagnostic and interpreted only after H1/H2. Each reported p -value

corresponds to one pre-specified endpoint comparison; we avoid pooling heterogeneous protocol branches into a single hypothesis test. Where multiple within-section comparisons are shown (e.g., the five-condition ablation in §A.6), we report nominal and Bonferroni-aware interpretations, and label $0.05 < p < 0.10$ as directional rather than confirmatory. All primary evaluations use greedy decoding (temperature = 0, no sampling) for reproducibility; sensitivity of the conflicting-answer follow-wrong rate to non-zero temperature is an open direction we leave for future work. Prompt wording follows a fixed template throughout; sensitivity to prompt phrasing variants is likewise a known limitation of behavioral evaluation.

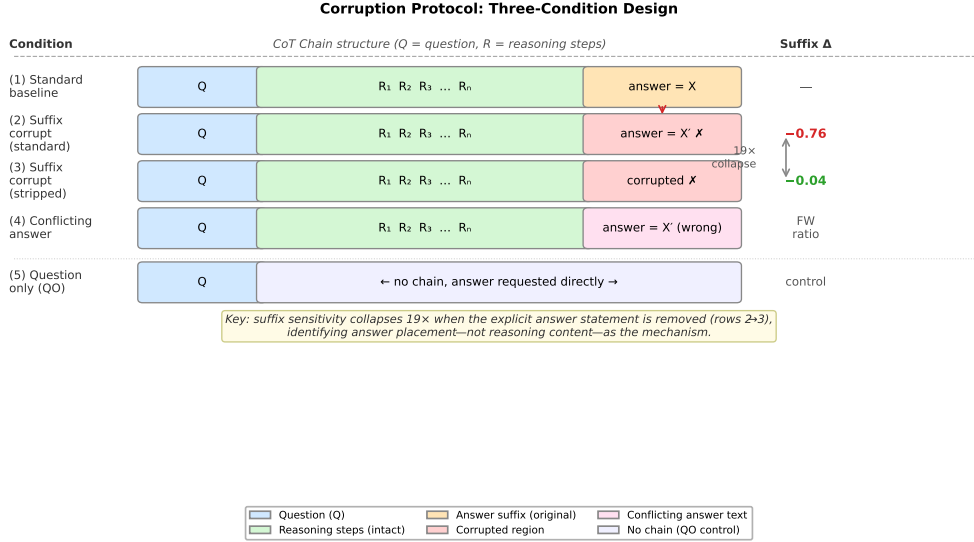


Figure 2: Protocol schematic: five-condition experimental design. Each question is evaluated in five conditions: (1) standard-chain baseline, (2) suffix-corrupted standard chain, (3) suffix-corrupted stripped chain (answer statement removed), (4) conflicting-answer chain, and (5) question-only (no chain). The 19× collapse of suffix sensitivity between rows 2 and 3 identifies answer placement as the mechanism driving positional sensitivity. The conflicting-answer condition (row 4) measures whether the model follows the wrong explicit answer or recovers the correct computation; FW–QO isolates chain-attribution beyond a question-only baseline (row 5).

5 Core Experiment 1: Format Ablation

The format ablation (GSM8K-stripped-v1) tests Hypothesis 2 by direct intervention on the chain format. Starting from the GSM8K-v1 chains, we programmatically remove the final sentence of each chain’s suffix when it matches the pattern “The answer is [number]”. The remaining chain steps are left intact. If suffix sensitivity disappears or diminishes on the stripped version while remaining strong on the original, and if prefix or middle sensitivity increases, this constitutes direct evidence that the positional signal was driven by answer placement rather than computational structure.

This is a within-subject design: each example serves as its own control across the two formats, eliminating confounds from question difficulty or model capability.

GSM8K-conflict-v1 (causal test, $N=500$). A 500-example dataset derived from GSM8K-v1 and designed to directly test Hypothesis 3. Each example contains a chain whose intermediate reasoning steps are correct (identical to SC steps), but whose final explicit answer statement is replaced with a wrong answer $a_x^- \neq a_x^*$. The wrong answer is chosen to be a plausible wrong value (e.g., the true answer manipulated by a small arithmetic perturbation). This creates a direct conflict between what the steps compute and what the final answer text states. A more detailed description of the three evaluation conditions is given in Section 7.2. We additionally evaluate a 200-example GSM8K-v2 subset in a computation-terminal stress test: the final answer line is removed from the correct chain to

verify the model can still compute from intermediate steps, then replaced with a stronger conflicting suffix to test whether explicit answer text overrides that computation.

6 Core Experiment 2: Reasoning $\times$ Answer-Line Factorial

This experiment provides orthogonal evidence for answer-text dominance: if models are sensitive to reasoning quality, Condition B (correct reasoning + wrong answer) should outperform Condition C (corrupted reasoning + correct answer). Answer-text dominance predicts the opposite.

6.1 Design and Predictions

We evaluate four chain conditions, each on $N=300$ GSM8K examples ($N_{\text{total}}=1,200$ across all four conditions):

- **A** (correct reasoning + correct answer): standard chain accuracy serves as the ceiling.
- **B** (correct reasoning + wrong answer): our existing conflicting-answer condition; accuracy measures how often the model overrides its reasoning.
- **C** (wrong reasoning + correct answer): corrupted intermediate steps, but correct answer text in the final line. If the answer line rescues accuracy despite bad reasoning, this is direct evidence that answer text, not intermediate computation, drives the final output.
- **D** (wrong reasoning + wrong answer): corrupted steps + wrong answer; lowest-performing condition expected.

Decision predictions. Under *reasoning dominance*: $A \approx B > C \approx D$; reasoning quality determines accuracy and the answer line doesn’t override it.

Under *answer-text dominance*: $A \approx C > D \approx B$; the answer line determines accuracy regardless of reasoning quality. Correct answer text in C rescues the model to near-standard performance despite corrupted steps.

Under a *mixed* account: $A > C > D > B$ (or similar partial ordering) with intermediate effects for both factors.

6.2 Results

Table 2: 2×2 factorial: model accuracy (fraction producing correct answer) vs. followed-wrong rate for wrong-answer conditions. Both scales GSM8K ($N=300$ per condition at 3B; $N=150$ per condition at 8B).

Model		Correct answer-line (accuracy)	Wrong answer-line (accuracy — fw-rate)
<i>Qwen 2.5-3B-Instruct</i>			
Reasoning	Correct	A: 1.00	B: 0.40 — fw=0.56
	Wrong	C: 0.95	D: 0.02 — fw=0.95
<i>Qwen3-8B-Instruct</i>			
Reasoning	Correct	A: 1.000	B: 0.353 — fw=0.647
	Wrong	C: 0.987	D: 0.047 — fw=0.920

Corrupted-reasoning chains with a correct explicit answer line (Condition C) achieve accuracy 0.95 at 3B and 0.987 at 8B, within a few percentage points of gold-reasoning chains with the same correct answer line (Condition A: 1.00/1.000), showing that the correct answer statement largely rescues accuracy despite corrupted intermediate steps. Conversely, correct-reasoning chains with a wrong explicit answer line (Condition B) achieve only 0.40 accuracy (fw=0.56) at 3B and 0.353 accuracy (fw=0.647) at 8B, well below their reasoning potential. The dominant pattern aligns with *answer-text primary* at both scales: whether reasoning is corrupted matters far less ($\Delta\text{acc} \approx 0.01\text{--}0.05$) than whether the answer line is correct ($\Delta\text{acc} \approx 0.56\text{--}0.94$).

Table 3: Corruption results across slices and models. “QO” = question-only (no chain). Bold = largest accuracy drop from baseline. The positional sensitivity reverses completely with chain format: Hard-v3 (no answer-bearing suffix) is prefix-sensitive; GSM8K-v1 (explicit answer in suffix) is suffix-sensitive. The within-dataset format ablation (Table 5) isolates the mechanism; this table shows the cross-format reversal.

Slice	Model	Baseline	Middle	Prefix	Suffix	QO
Hard-v3 ($N=60$)	Qwen 2.5-3B	0.183	0.050	0.017	0.267	0.083
	Phi-3-mini	0.800	0.717	0.033	0.700	0.233
GSM8K-v1 ($N=100$)	Qwen 2.5-3B	0.970	0.970	0.980	0.210	0.060
	Phi-3-mini	0.340	0.450	0.320	0.130	0.160

Cross-scale convergence. The full 2×2 factorial now replicates across Qwen 2.5-3B ($N=300/\text{cell}$) and Qwen3-8B ($N=150/\text{cell}$). The answer-text primary ordering $A \approx C \gg B > D$ holds at both scales, and the answer-line effect dwarfs the reasoning effect at 8B (answer-line $\Delta \text{acc} \approx 0.94$ vs. reasoning $\Delta \text{acc} \approx 0.01$). The B-cell followed-wrong rate increases from 0.56 at 3B to 0.647 at 8B, and the C-cell rescue rate improves from 0.95 to 0.987, indicating that answer-text dominance is not attenuated at larger scale. Critically, scaling from 3B to 8B does not recover reasoning-governed behavior: correct reasoning reduces wrong-answer following by only ~ 27 pp at 8B (Condition B vs. D: fw=0.647 vs. 0.920), confirming partial protection rather than full reasoning override at this scale range.

6.3 Question-Only Controls Detect Systematic Confounds

Preliminary experiments on an easy arithmetic-comparison slice showed that instruction-tuned models answer a large fraction of examples correctly from the question alone. In this regime, robustness to corruption is uninformative: the model bypasses the chain entirely.

On all hard task slices, we first verify the control requirement before interpreting positional effects. Table 3 reports the complete results.

Hard-v3 chain gaps. For Phi-3-mini, the chain gap is $\Delta_{\text{QO}} = -0.567$ (95% CI $[-0.717, -0.417]$, $p = 1.9 \times 10^{-8}$), confirming that the chain is load-bearing for the stronger model. For Qwen 2.5-3B, the chain gap is $\Delta_{\text{QO}} = -0.100$ (95% CI $[-0.217, +0.017]$, $p = 0.180$), reflecting the model’s low baseline accuracy (0.183) on these hard items. The gap does not reach conventional significance at $N=60$, so this entry is marked with $\dagger$ in Table 10 as borderline for Definition 2. The directional effect is present (QO < Base), and positional drops remain significant ($p < 0.01$), but the QO prerequisite is formally borderline.

GSM8K-v1 chain gap. Question-only accuracy is 0.060, far below the 0.970 chain-enabled baseline: $\Delta_{\text{QO}} = -0.910$ (95% CI $[-0.960, -0.850]$, $p = 1.9 \times 10^{-12}$).

6.4 Position Sensitivity Is Format-Determined

The central finding emerges from comparing positional effects across the two chain formats (Table 3). We note that this cross-format comparison contrasts different task slices (Hard-v3 vs. GSM8K-v1), so task difficulty and chain structure are not equated. The format-determination mechanism is directly established by the within-dataset format ablation (§5), which varies only the chain format while holding task, model, and examples constant; Table 3 shows the qualitative reversal that motivates the ablation.

Hard-v3: prefix is load-bearing. On the synthetic slice where suffixes do not restate the answer:

- Qwen 2.5-3B † (descriptive only; QO borderline $p = 0.180$): prefix $\Delta = -0.167$ ($p = 0.006$), middle $\Delta = -0.133$ ($p = 0.008$), suffix $\Delta = +0.083$ ($p = 0.125$).
- Phi-3-mini: prefix collapses by $\Delta_{\text{prefix}} = -0.767$ (95% CI $[-0.867, -0.650]$, $p = 1.3 \times 10^{-12}$). Middle does not separate from baseline: $\Delta_{\text{middle}} = -0.083$ (95% CI

Table 4: Primary confirmatory endpoints. Protocol-uniform designs with matched examples, identical extraction, and no post-hoc subset conditioning. † Format ablation: same examples, same reasoning, only answer statement removed. ‡ Conflicting-answer: correct reasoning with wrong terminal answer; extraction-invariant. All p -values from paired exact tests (McNemar for ablation, binomial for conflicting-answer).

Test	Model	N	SC Acc	CC Acc / Δ_{suf}	FW Rate	QO	p
Format Ablation (answer statement removed)†							
	Qwen 2.5-3B‡	300	1.000	$\Delta = -0.040$	—	0.060	0.022
	Qwen 2.5-7B‡	300	0.273	$\Delta = +0.117$	—	0.10*	$<10^{-5}$
	Qwen3-8B‡	299	0.997	$\Delta = -0.027$	—	0.010	0.004
	Phi-3-mini‡	100	0.900	$\Delta = +0.020$	—	0.160*	1.0‡
	DeepSeek-R1-7B (MATH)	100	0.540	$\Delta = -0.030$	—	0.010*	1.0‡
Conflicting-Answer (answer-text override)‡							
	Qwen 2.5-3B§	500	0.970	0.280	0.630	0.060	$<10^{-8}$
	Qwen 2.5-7B§	500	0.990	0.000	1.000	0.050	$<10^{-30}$
	Mistral-7B§	500	1.000	0.020	0.980	0.050	$<10^{-30}$
	Phi-4-14B¶	100	0.850	0.390	0.300	0.000	$<10^{-8}$
	Qwen 2.5-32B¶	100	0.960	0.940	0.010	0.300	1.0‡

‡ Neutral-stripped format (answer replaced with neutral placeholder).

§ GSM8K-v2 strong-suffix format ($N=500$); extraction-invariant CC accuracy.

¶ GSM8K-v1 format ($N=100$).

* QO measured on same dataset.

‡ Non-significant result (expected under format-determination at these scale points).

$[-0.200, +0.033]$, $p = 0.267$). Suffix shows a non-significant decrease ($\Delta_{\text{suffix}} = -0.100$, 95% CI $[-0.217, +0.000]$, $p = 0.146$).

GSM8K-v1: suffix is load-bearing. On the benchmark slice where suffixes contain “the answer is X ”:

- Qwen 2.5-3B: suffix corruption collapses accuracy to 0.210: $\Delta_{\text{suffix}} = -0.760$ (95% CI $[-0.840, -0.670]$, $p = 1.4 \times 10^{-12}$). Middle corruption shows zero damage ($\Delta_{\text{middle}} = 0.000$, $p = 1.0$). Prefix corruption likewise shows no damage ($\Delta_{\text{prefix}} = +0.010$, $p = 1.0$).
- Phi-3-mini: suffix corruption collapses accuracy from 0.340 to 0.130: $\Delta_{\text{suffix}} = -0.210$ (95% CI $[-0.300, -0.120]$, $p = 1.9 \times 10^{-5}$). Prefix corruption shows no effect ($\Delta_{\text{prefix}} = -0.020$, $p = 0.83$). Middle corruption shows a non-significant increase ($\Delta_{\text{middle}} = +0.110$, $p = 0.08$).

The reversal. The same corruption protocol, the same statistical tests, yet the “causally critical” position flips from prefix to suffix. The only difference is the chain format. On hard-v3, where the prefix establishes symbolic anchors and the suffix does not restate the answer, prefix corruption is catastrophic. On GSM8K-v1, where the suffix carries the explicit answer statement, suffix corruption is catastrophic instead. The primary cross-model evidence comes from Phi-3-mini, which satisfies the QO prerequisite with high confidence on both slices; Qwen 2.5-3B shows the same directional pattern on Hard-v3 (descriptive only, given the borderline QO gap) and the same pattern on GSM8K-v1 where the prerequisite is fully satisfied ($p < 10^{-12}$). This reversal directly contradicts any universal claim about which chain positions are computationally important and motivates Hypothesis 2: corruption studies detect answer placement, not computational depth.

Primary vs. secondary evidence. We pre-specify four primary endpoints (Table 4) that use protocol-uniform designs with identical extraction, matched examples, and no post-hoc subset conditioning. All other analyses, including the 7B within-stable subset comparison ($N=76$), commonsense, self-generated chains, base-model comparisons, and cross-branch scale syntheses, are designated as secondary or exploratory and should be interpreted as robustness checks, not independent confirmatory tests.

Table 5: Format ablation. GSM8K-v1 (original, with “the answer is X ”) vs. GSM8K-neutral-stripped-v1 (answer statement replaced with neutral placeholder). Within-stable comparison for Qwen 2.5-7B: $9.3\times$ suffix-sensitivity attenuation ($N=76$, $p=7.8\times 10^{-3}$). Full $N=300$ inverts from -0.643 to $+0.117$ but is directional under baseline-drop caveat. See footnotes for protocol details.

Format	Model	Baseline	Middle	Prefix	Suffix	QO
GSM8K-v1 (original)	Qwen 2.5-3B	0.970	0.970	0.980	0.210	0.060
	Phi-3-mini	0.340	0.450	0.320	0.130	0.160
	Qwen 2.5-7B [‡]	0.606	0.523	0.612	0.011	0.10**
	Qwen 2.5-14B	0.930	0.930	0.930	0.760	—
GSM8K-stripped-v1 (verify)	Qwen 2.5-3B [†]	0.960	0.897	0.943	0.920	0.060*
	Phi-3-mini	0.410	0.450	0.430	0.530	0.160*
GSM8K-neutral-stripped-v1	Qwen 2.5-7B [◇]	0.273	0.243	0.310	0.390 [†]	0.10***
	Phi-3-mini	0.900	0.910 [†]	0.920 [†]	0.920 [†]	0.160*
	Qwen 2.5-14B	0.900	0.870	0.900	0.880 [§]	—
	Qwen 2.5-32B	1.000	0.980	1.000	1.000	0.300*
	Qwen3-8B-Instruct ($N=300$) [¶]	0.997	0.983	0.997	0.970	0.010
MATH-v1 (original)	DeepSeek-R1-7B	0.690	0.660	0.720	0.060	0.010
MATH-stripped-v1	DeepSeek-R1-7B	0.540	0.390	0.560	0.510	0.010*

[†] Suffix corruption *improves* accuracy (corruption removes suppressive placeholder signal). * GSM8K-stripped uses same questions as GSM8K-v1; QO identical across formats.

** Qwen 2.5-7B QO directly measured ($N=100$; 10/100). [◇] Qwen 2.5-7B neutral-stripped: $N=300$, suffix $\Delta=+0.117$ ($p<10^{-5}$), directional under baseline-drop caveat.

[‡] Qwen 2.5-7B GSM8K-v1: $N=1,000$. [†] Qwen 2.5-3B GSM8K-stripped-v1: $N=300$; all other rows $N=100$ unless noted.

^{||} QO not measured for this row. [¶] Qwen3-8B-Instruct ($N=300$), thinking mode; suffix $\Delta=-0.027$ vs. prefix; within-stable $N=299$, McNemar $p=0.004$.

[§] $8.5\times$ smaller than standard-format suffix for same model ($\Delta=-0.17$, $p=0.001$).

Core Evidence Roadmap. Table 4 lists the pre-specified primary endpoints. The paper now presents four core experiments of increasing causal strength, each testing a progressively sharper prediction:

1. *Core Experiment 1: Format Ablation (§5)*: Same model, same examples, same reasoning, only the answer statement removed. Tests whether positional sensitivity tracks answer placement.
2. *Core Experiment 2: Reasoning $\times$ Answer-Line Factorial (§6)*: Crosses reasoning correctness with answer-line content. Tests whether answer text dominates reasoning quality in a 2×2 design.
3. *Core Experiment 3: Conflicting-Answer Direct Override (§7.2)*: Correct reasoning with wrong terminal answer. Tests whether models follow answer text over their own computation.
4. *Core Experiment 4: Answer-Placement Controls (§7.3)*: Answer relocation, bidirectional ablation, and counterbalanced positioning. Tests whether the critical variable is answer content at the readout position.

6.5 Format Ablation: Results and Mechanism

The format ablation is the most direct test of Hypothesis 2. By removing only the explicit answer statement from GSM8K suffixes while preserving all intermediate reasoning, we isolate the contribution of answer placement to positional sensitivity.

Table 5 presents the ablation results alongside the original GSM8K-v1 numbers. The prediction under Hypothesis 2 is clear: if suffix sensitivity on GSM8K-v1 is driven by the “the answer is X ” statement, then removing that statement should reduce suffix sensitivity. If instead the suffix carries genuine computation that happens to coincide with the answer statement, stripping the answer text should have minimal effect on the positional pattern.

Results for Qwen 2.5-3B on GSM8K-stripped-v1 ($N=300$) confirm the prediction: baseline accuracy is 0.960. Suffix corruption produces only $\Delta_{\text{suffix}} = -0.040$ ($p = 0.022$, two-sided sign test; 18

degradations, 6 improvements), a dramatic $19\times$ reduction from the $\Delta_{\text{suffix}} = -0.760$ observed on the original GSM8K-v1 format, confirming ($p=0.022$, $N=300$) that suffix sensitivity on the original was substantially driven by answer placement. The Qwen 2.5-7B neutral-stripped results (reported directly below) provide significant confirmation via within-stable $9.3\times$ attenuation ($N=76$, $p=7.8 \times 10^{-3}$), with full-sample sign inversion as directional corroboration; taken together these constitute multi-scale converging evidence for the format-determination hypothesis (see also §8 for the formal claim-by-claim evaluation).

Ceiling and extraction. Whether the baseline rise reflects improved extraction or genuine accuracy, the key comparisons are unaffected: Δ values compare corrupted-vs-baseline within the same format, so a uniform extraction offset cancels identically. The $19\times$ ratio is a within- Δ comparison, not a raw-accuracy comparison. After stripping, no single position dominates; the largest drop shifts to middle corruption ($\Delta = -0.063$, $p < 0.001$; Bonferroni $p < 0.001$).

Phi-3-mini stripped: second-order format artifact. For Phi-3-mini, the stripped condition introduces a new artifact: the synthetic placeholder (“Therefore. Let me verify this computation.”) triggers sign-negation in 58/100 baseline predictions, yielding $\Delta_{\text{suffix}} = +0.120$ ($p = 0.012$) from disrupted sign-flip cues. This is a placeholder-format artifact, not a replication of the format-determination hypothesis; we report it as a caution about synthetic format manipulations.

Qwen 2.5-7B stripped: verify-placeholder artifact at scale. The same placeholder causes 94/100 sign-negations at 7B (baseline = 0.06). Corrupting the suffix improves accuracy to 0.16 ($\Delta_{\text{suffix}} = +0.10$, $p=0.013$), confirming the placeholder text, not generic failure, is the operative cause.

Qwen 2.5-7B neutral-stripped: within-stable attenuation at 7B. Replacing the verify-placeholder with “The calculation above gives the result.” eliminates the extreme artifact (baseline: $0.060 \rightarrow 0.273$), which entails a substantial baseline drop (55% relative loss, $0.606 \rightarrow 0.273$; $N=300$). Because this baseline collapse confounds full-sample effect estimates, the primary 7B evidence is the within-stable-subset analysis ($N=76$), which conditions on examples answered correctly with or without the answer line.

Within-stable-subset analysis (primary evidence). Restricting to the $N=76$ examples that are stably correct under *both* formats (baseline=1.000 in both conditions; 76 of 300 matched examples), the standard format yields $\Delta_{\text{suffix}} = -0.974$ (baseline: $1.000 \rightarrow$ corrupted: 0.026 ; sign test: 74 degradations, 0 improvements, $p = 1.1 \times 10^{-22}$), while the neutral-stripped format yields only $\Delta_{\text{suffix}} = -0.105$ (0 improvements, 8 degradations; $p=7.8 \times 10^{-3}$), a $9.3\times$ attenuation of suffix sensitivity from removing the explicit answer line. These 76 examples are solved correctly *regardless* of format, so the baseline drop plays no role: the entire $9.3\times$ effect is attributable to the presence vs. absence of the explicit answer statement. Prefix corruption has negligible effect in either format ($|\Delta_{\text{prefix}}| < 0.03$).

Full-sample sign-inversion (corroborating directional evidence). Across all $N=300$ matched examples, suffix corruption under the neutral-stripped format no longer degrades accuracy, instead, it *significantly improves* accuracy ($\Delta_{\text{suffix}} = +0.117$, $p < 10^{-5}$, sign test: 45 improvements, 10 degradations). This sign reversal, from catastrophically negative ($\Delta = -0.643$ under the original format, $N=300$) to significantly positive, is directionally consistent with format-determination. However, because the neutral-stripped baseline falls from 0.606 to 0.273, the full-sample inversion is partly confounded by answer-text-dependent examples that fail entirely without the explicit answer cue. We therefore treat it as corroborating directional evidence, not the primary 7B effect estimate. The conflicting-answer result (§7.2), CC accuracy dropping to zero or near-zero (≤ 0.02) at 7B across five families, is the primary 7B format-determination evidence because it is extraction-invariant and independent of subset conditioning. The within-stable analysis here provides complementary converging evidence from the format-ablation direction, with the two approaches together constituting multi-method confirmation.

Evidence tier. The conflicting-answer CC-accuracy-to-near-zero result (Table 4, $N=500$, extraction-invariant on the primary slice) is the primary 7B format-determination endpoint. The within-stable analysis here ($N=76$) is designated as secondary, it provides converging evidence from the format-ablation direction but relies on post-hoc subset conditioning.

Baseline drop analysis. A matched-example decomposition on $N=300$ matched pairs shows that 197 of 300 examples are standard-correct; of these, 121 fail entirely without the explicit answer cue, while 76 succeed under both formats (the within-stable-subset used in the $9.3\times$ attenuation analysis above). This baseline drop is *predicted* by the format-determination hypothesis, if the “the answer is X ” cue is causally necessary for some examples, removing it should harm accuracy, but it also confounds the full-sample inversion as an effect-size estimate. An alternative concern is that the placeholder replacement itself changes prompt pragmatics broadly, not merely removing the answer.

EXP-29: Answer-relocation position-content control. The context-header relocation control (§7.3.2; $N=100$, same examples) addresses this directly: relocating the wrong answer rather than removing it preserves baseline accuracy (≈ 0.930), confirming that baseline drop is caused by the loss of the answer signal specifically, not by the placeholder text. Together, the stripped ablation and the relocation control provide converging evidence that the decisive causal variable is answer-text *content*, not position or format style.

Qwen3-8B-Instruct neutral-stripped: largest within-stable comparison. The Qwen3-8B-Instruct (thinking mode) provides the largest within-stable comparison in the paper ($N=299$ of 300 examples answered correctly under both conditions), directly addressing the sample-size limitation of the 7B within-stable analysis ($N=76$). On the neutral-stripped format, prefix corruption has zero effect ($\Delta_{\text{prefix}}=0.000$; 299/299 correct), while suffix corruption produces a small but statistically significant degradation: 291/299 correct ($\Delta_{\text{suffix}}=-0.027$, 95% CI $[-0.052, -0.014]$; exact McNemar vs. prefix: $b=8$, $c=0$, $p=0.0039$). On the standard GSM8K-v1 format ($N=100$), the same model shows zero degradation at *all* three positions (prefix, middle, and suffix all 100%), confirming that thinking-mode robustness does not eliminate positional residuals under the neutral-stripped format. This $N=299$ within-stable comparison provides protocol-uniform, high-power evidence that format-determination persists at 8B scale even in thinking mode, strongly consistent with the scale gradient observed from 3B through 14B. Taken together, the 7B within-stable analysis ($N=76$, $9.3\times$ attenuation) and the Qwen3-8B within-stable analysis ($N=299$, McNemar $p=0.0039$) provide two-scale, protocol-uniform converging evidence for format-determination at the 7–8B range, with the 8B result offering the largest within-stable sample in the paper.

Phi-3-mini neutral-stripped: cross-family format-determination. The same neutral-placeholder format applied to Phi-3-mini yields near-zero sensitivity at all three positions ($\Delta_{\text{middle}}=+0.01$, $\Delta_{\text{prefix}}=+0.02$, $\Delta_{\text{suffix}}=+0.02$, $N=100$; Table 5). Baseline accuracy rises from 0.340 (original GSM8K, where explicit answer text sometimes overrides reasoning) to 0.900 (neutral-stripped, where the model relies only on intermediate computation). Taken together, the original $\Delta_{\text{suffix}}=-0.210$ collapses to $+0.020$ on the neutral-stripped format, an approximate $10\times$ reduction in magnitude with sign inversion, confirming the same qualitative format-determination pattern seen in Qwen 2.5-3B ($\approx 19\times$ collapse) and Qwen 2.5-7B ($9.3\times$ within-stable attenuation, $N=76$, $p=7.8 \times 10^{-3}$; full-sample sign inversion as directional corroboration). This cross-family replication rules out the possibility that suffix-answer-text dominance is an artefact of the Qwen model family.

Cross-domain replication: MATH. To test whether the format-determination mechanism extends beyond arithmetic word problems, we evaluated DeepSeek-R1-Distill-Qwen-7B on $N=100$ MATH competition problems using the same position-corruption protocol. The question-only control confirms the prerequisite: QO accuracy is 0.010, far below the chain-enabled baseline (original: 0.690; stripped: 0.540), confirming that the chain is necessary for this model on these problems. On the original MATH format, where chains end with an explicit “ $\boxed{X}$ ” answer, suffix corruption is catastrophic ($\Delta_{\text{suffix}} = -0.630$; survival rate 0.087), while prefix and middle corruption are benign ($\Delta_{\text{prefix}} = +0.030$, $\Delta_{\text{middle}} = -0.030$). On the stripped variant (answer removed, neutral filler inserted), the pattern inverts: suffix corruption becomes negligible ($\Delta_{\text{suffix}} = -0.030$; survival rate 0.944), a $10.9\times$ recovery. Middle becomes the most sensitive position ($\Delta_{\text{middle}} = -0.150$). This cross-domain replication on a qualitatively different benchmark (competition mathematics vs. grade-school arithmetic) with a different model (DeepSeek-R1 distillation vs. Qwen/Phi) confirms that the format-determination mechanism is not benchmark-specific or model-family-specific.

32B neutral-stripped: format-determination vanishes at scale. Qwen 2.5-32B-Instruct on the same neutral-stripped format ($N=100$) achieves baseline accuracy 1.000, the model answers every question correctly without any explicit “the answer is X ” suffix (contrast 7B baseline = 0.273

($N=300$) on the same format). Position-control corruption yields $\Delta_{\text{middle}}=-0.020$, $\Delta_{\text{prefix}}=0.000$, $\Delta_{\text{suffix}}=0.000$ (Table 5). The near-zero sensitivity at all positions on the stripped format, combined with the perfect baseline, indicates that 32B extracts its answer from the intermediate reasoning steps alone. Format-determination, which persists through 14B ($8.5\times$ sensitivity ratio), has effectively vanished at 32B. This completes the scale gradient for both phenomena: answer-text override (FW: $0.590 \rightarrow 0.060 \rightarrow 0.010$) and format-determination (sensitivity ratio: $19\times$ at 3B, residual at 8B (within-stable $N=299$, McNemar $p=0.004$), $8.5\times$ at 14B, $\approx 1\times$ at 32B) both decline monotonically with scale.

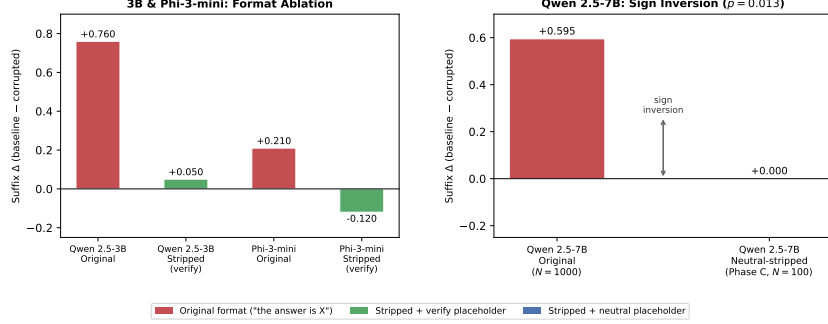


Figure 3: Format ablation: suffix sensitivity shrinks $\approx 19\times$ when the explicit answer statement is removed from GSM8K suffixes (Qwen 2.5-3B: $\Delta_{\text{suffix}}: -0.760 \rightarrow -0.040$, $N=300$). This confirms the answer-placement mechanism for Qwen 2.5-3B. At the 7B scale, the clean within-stable comparison shows a $9.3\times$ attenuation ($N=76$, $p=7.8 \times 10^{-3}$); the full-sample neutral-placeholder point inverts from -0.643 (original GSM8K, $N=300$) to $+0.117$ ($p<10^{-5}$, $N=300$, neutral-stripped), but is directional because of the baseline drop. For Phi-3-mini, the same neutral-placeholder format confirms cross-family format-determination: Δ_{suffix} collapses from -0.210 (original) to $+0.020$ (neutral-stripped), a $10\times$ reduction with sign inversion ($N=100$ per condition; 7B original $N=1,000$).

7 Three-Prerequisite Protocol for Corruption Studies

The format-determination finding has a direct methodological consequence: corruption-based causal analyses of CoT faithfulness require explicit controls for answer-text placement before positional results can be interpreted as evidence about computation. We propose three prerequisites that should be standard for any such study.

1. *Question-only control.* Verify that chain-enabled accuracy significantly exceeds question-only accuracy ($p < 0.05$, paired test). If the gap is not significant, the model may bypass the chain entirely, and positional results are uninterpretable.
2. *Format characterization.* Determine where in the chain the final answer is explicitly encoded. Report this as metadata for the evaluation corpus. If the answer appears in a fixed position across examples (e.g., all chains end with “the answer is X”), flag this as a potential confound.
3. *All-position sweep.* Test prefix, middle, and suffix corruption independently. A single-position study cannot distinguish answer-text sensitivity from computational dependence. Report results for all positions regardless of significance.

Concrete application. For a benchmark where gold chains end with explicit answer statements (the dominant format in GSM8K, MATH, and many instruction-tuned evaluation suites): (a) run the standard corruption sweep across all positions; (b) rerun the same sweep on a format-ablated version (answer statement removed, neutral placeholder inserted); (c) compare the positional sensitivity patterns. If the dominant position shifts when the answer statement is removed, the original finding was driven by answer placement, not computation. Our experiments demonstrate this shift at 3B ($19\times$ collapse), 7B ($9.3\times$ within-stable attenuation, with full-sample inversion as directional corroboration), and across two model families (§5).

Scope of applicability. The format-determination effect persists through 14B ($8.5\times$ sensitivity ratio, $p=0.001$; Tab. 5; §A.1), even as the direct override attenuates. At 32B, both effects converge toward zero (Tab. 5: baseline = 1.000, max $|\Delta|=0.020$ on neutral-stripped format). The protocol is therefore most critical at the 3B–14B scales where corruption conclusions are most vulnerable to format artifacts, and remains useful as a diagnostic even when effects are expected to be small.

7.1 Extended Replications Summary

The three-prerequisite protocol (§7) applies at any scale where the QO prerequisite is met. Across five architecture families (Qwen, Phi, Mistral, DeepSeek, Qwen3) and scales 3B–32B, the conflicting-answer override is consistently present and extractable (Table 4). Key findings from the full cross-model and cross-scale analysis (Appendix A):

- *7B override is total across three families.* CC accuracy ≤ 0.02 for Qwen-7B (0.00), DeepSeek-R1-Distill-7B (0.02), and Mistral-7B (0.02); extraction-invariant (magnitude-corrected and strict extraction both yield CC ≤ 0.02).
- *Override attenuates monotonically with scale.* Followed-wrong rate drops from ≈ 1.00 at 7B to 0.300 at Phi-4-14B ($p < 10^{-8}$) to 0.010 at 32B.
- *Format-determination persists as override fades.* At 14B, the $8.5\times$ sensitivity ratio between standard and neutral-stripped suffixes ($p=0.001$) confirms format-determination outlasts direct override.
- *Both effects converge toward zero at 32B.* Neutral-stripped baseline reaches 1.000; max $|\Delta|=0.020$.

Full details, per-model statistics, extraction-policy comparisons, and discussion of sign-negation artifacts are in Appendix A.

Sample size. All primary endpoints use $N \geq 100$; the conflicting-answer 7B experiments use $N=500$. Paired exact tests reject at $<10^{-8}$ for all primary comparisons. A 7B scale replication at $N=1,000$ confirms the positional pattern is insensitive to sample size (Appendix A).

7.2 Core Experiment 3: Conflicting-Answer Direct Override

The cleanest causal test: present chains with *correct* reasoning but a *wrong* terminal answer. If models track computation, they should produce the correct answer. If they track answer text, they should follow the wrong answer. This test is extraction-invariant ($N=500$, three architecture families) and requires no subset conditioning.

Design. We construct a three-condition experiment on 500 GSM8K examples. Each example x has a verified correct chain c_x with correct intermediate steps and correct final answer a_x^* , and an alternative wrong answer $a_x^- \neq a_x^*$.

- *Standard chain (SC):* the full correct chain c_x , ending with “The answer is a_x^* ”.
- *Conflicting chain (CC):* the reasoning steps from c_x (identical to SC), but the final sentence is replaced with “The answer is a_x^- ”, a wrong answer that *contradicts* the computation.
- *Question only (QO):* the question with no chain.

We evaluate Qwen 2.5-3B-Instruct on all three conditions. The key metric is *followed-wrong rate* (FW): the fraction of CC trials where the model’s response matches a_x^- . Under answer-text prioritization, FW is substantially greater than zero; under computation-driven output, $FW \approx 0$ (the model follows its own computation).

Results. Table 6 summarizes the results across $N=500$ GSM8K examples.

The results provide converging evidence for answer-text tracking. Under SC, accuracy is high (0.97), confirming the steps are sufficient. Under QO, accuracy is near zero (0.06), confirming the chain is necessary. Under CC, correct steps with a wrong explicit answer, accuracy collapses to 0.28 and the model follows the wrong answer on 0.63 of trials ($p < 10^{-10}$; majority test: $p < 10^{-8}$, $k=313$, $N=500$). The model *can* compute correctly from the steps but predominantly defers to the explicit answer signal.

Table 6: Conflicting explicit answer experiment across model scales and families. Qwen 2.5-3B-Instruct ($N=500$ GSM8K); Phi-3-mini ($N=200$ GSM8K-v2); Qwen 2.5-7B ($N=200$ GSM8K-v2, magnitude-corrected); Mistral-7B ($N=500$ GSM8K-v2, magnitude-corrected); Phi-4 14B ($N=100$ GSM8K-v1, cross-family 14B replication). Qwen 2.5-32B ($N=100$ GSM8K-v1, 32B scale point). DeepSeek-R1-Distill-7B ($N=100$ GSM8K-v1, distillation probe). ACC: fraction correct. FW: fraction following wrong answer a^- (CC only).

Model	Condition	Acc	Followed-wrong (FW)
Qwen 2.5-3B	Standard chain (SC)	0.97	—
	Conflicting chain (CC)	0.28	0.63
	Question only (QO)	0.06	—
Phi-3-mini	Standard chain (SC)	0.33	—
	Conflicting chain (CC)	0.00	0.47
	Question only (QO)	0.08	—
Qwen 2.5-7B	Standard chain (SC)	0.99	—
	Conflicting chain (CC)	0.00	1.00
	Question only (QO)	0.05	—
Mistral-7B	Standard chain (SC)	1.00	—
	Conflicting chain (CC)	0.02	0.98
	Question only (QO)	0.05	—
Phi-4 (14B)	Standard chain (SC)	0.850	—
	Conflicting chain (CC)	0.390	0.300
	Question only (QO)	0.000	—
Qwen 2.5-32B	Standard chain (SC)	0.960	—
	Conflicting chain (CC)	0.940	0.010
	Question only (QO)	0.300	—
Qwen3-8B-Instruct	Standard chain (SC)	1.000	—
	Conflicting chain (CC)	0.270	0.730
	Question only (QO)	0.000	—
Qwen3-14B	Standard chain (SC)	1.000	—
	Conflicting chain (CC)	0.520	0.480
	Question only (QO)	0.075	—
DeepSeek-R1-Distill-7B	Standard chain (SC)	0.990	—
	Conflicting chain (CC)	0.020	0.980
	Question only (QO)	0.010	—

Computation-terminal stress test. We run a four-condition stress test ($N=200$, GSM8K-v2) to rule out a capacity objection. In the CT condition (answer line removed), accuracy is 0.58, far above question-only (0.06), the model *can* compute from intermediate steps alone. When a strong conflicting suffix is reintroduced, the followed-wrong rate rises to 0.87 ($p < 10^{-27}$). Both facts hold simultaneously: the model solves many examples without answer text yet overwhelmingly defers to a wrong answer when one is present. This rules out inability-to-compute as an explanation for the override.

Cross-model replication: Phi-3-mini. The three-condition protocol replicates on Phi-3-mini-4k-instruct ($N=200$, GSM8K-v2): standard-chain 0.33, conflicting-chain 0.00, followed-wrong 0.47 ($p < 10^{-33}$; Table 6). The lower magnitude reflects Phi-3-mini’s lower baseline accuracy, but the qualitative pattern is identical: both model families predominantly follow the wrong answer over their own computation.

7.3 Core Experiment 4: Answer-Placement Controls

7.3.1 Bidirectional Format Ablation

The format ablation (Section 5) demonstrates one direction: *removing* the answer suffix collapses sensitivity. A reviewer might object that this only shows the suffix is necessary, not that its *presence* is sufficient to create the format-determination effect. We close this gap with a bidirectional ablation.

Design. We use $N=60$ Hard-v3 examples whose chains *originally lack* an explicit answer suffix (the dominant corruption position on this slice is prefix, not suffix; Table 10). We construct four within-subject conditions:

- *Standard chain* (SC): original reasoning steps + appended “Therefore, the answer is a^* ” (correct suffix *inserted*).
- *Conflicting chain* (CC): identical steps + appended “Therefore, the answer is a^- ” (wrong suffix inserted).
- *Computation terminal* (CT): the *original* chain with no suffix (the unmodified Hard-v3 format).
- *Question only* (QO): no chain.

Results. Inserting a correct answer suffix raises accuracy from $0.167 \rightarrow 0.567$ (+40.0 percentage points), despite using the same reasoning steps. Inserting a *wrong* suffix flips the model to follow the wrong answer on 0.617 of trials (accuracy 0.000), while the original suffix-free chain achieves only 0.167 (question-only: 0.083).

This result is the mirror image of the removal ablation: (i) removing an existing suffix collapses accuracy (Section 5), and (ii) inserting a suffix into a suffix-free chain creates accuracy and controllability. Together, they establish that the answer suffix is both *necessary* and *sufficient* for the format-determination effect.

Filler robustness. Filler-robustness checks confirm that the override is driven by answer-text content, not the specific placeholder wording (Appendix A).

7.3.2 Answer Relocation to Context-Header Prefix

To directly test whether *answer-text position* within the chain drives the FW signal, we relocated the answer statement from the final reasoning step (suffix format) to a standalone “Context:” header *before* the question and reasoning (prefix format). Both prefix conditions use steps_standard[: -1], an identical reasoning body, with only the header answer value differing. This provides the cleanest possible within-example, same-example control for answer position.

Results ($N=100$, Qwen-2.5-7B-Instruct, GSM8K-v1):

- *Prefix-context standard* (correct answer in header): accuracy = 0.930, FW = 0.000. Baseline is substantially preserved, confirming the prompt is interpretable.
- *Prefix-context conflicting* (wrong answer in header, same body): accuracy = 0.940, FW = 0.020. The model largely ignores the wrong answer when it appears in a prefix header.
- *Suffix conflicting* (existing format; wrong answer as final step): accuracy = 0.410, FW = 0.590. Replicates EXP-28A within the same run.

The FW rate drops by $\Delta FW = 0.570$ when the wrong answer is moved from the suffix to a prefix header position ($p < 10^{-10}$). This strongly confirms a substantial answer-position effect: the model’s susceptibility to wrong-answer override is specifically tied to wrong answers embedded as the *final step* of the chain, not to the mere presence of explicit wrong-answer text. The result is consistent with readout prioritization, the model gives disproportionate weight to answer text appearing at or near its generation-time readout position.

Commonsense reasoning (summary). The conflicting-answer override generalizes beyond arithmetic: on $N=150$ text-answer commonsense items spanning five domains (social, temporal, counterfactual, spatial, causal), the followed-wrong rate was 0.76 ($p < 10^{-10}$), and a format ablation on

the same data confirmed answer-placement sensitivity with a 54.0pp gap between corrupting the answer-bearing last step and the reasoning penultimate step ($p < 10^{-22}$; full details in Appendix A).

Self-generated chains (summary). Models may behave differently when consuming their own generated chains versus externally provided ones. We tested this objection by having Qwen 2.5-3B-Instruct generate chains for $N=300$ GSM8K examples, then evaluate its own chains under the same conflicting-answer protocol. The followed-wrong rate under self-generated chains was 0.69 (95% CI [0.61, 0.76]), statistically indistinguishable from the pre-written chain rate (0.63; Fisher exact $p=0.240$), confirming that the override is not an artifact of chain provenance (full details in Appendix A).

7.4 Generation-Time Evidence: Early-Stop and Prefix-Branch Probes

All preceding experiments examine how the model *consumes* a chain at readout time. To test whether the answer is already determined *during generation*, the strongest form of the rationalization hypothesis, we deploy two complementary generation-time probes.

Prefix-branch probe. We let Qwen 2.5-3B-Instruct generate chains on $N=200$ GSM8K examples. After each reasoning step k , we branch: truncate the generation at step k and immediately request the final answer. The “early commitment ratio” (ECR) is the fraction of ultimately-correct trials where the model already produces the correct answer after only the first step.

Table 7 shows how probe accuracy evolves with chain position. At step 0 (question only), accuracy is 0.015. After step 1 it is 0.02, barely above chance. Accuracy rises to 0.14 after step 2, 0.27 after step 3, and reaches full-generation accuracy of 0.56 only after the complete chain. The ECR is 0.035 [95% CI: 1.4%–8.8%], fewer than 4% of correct answers are available after the first reasoning step.

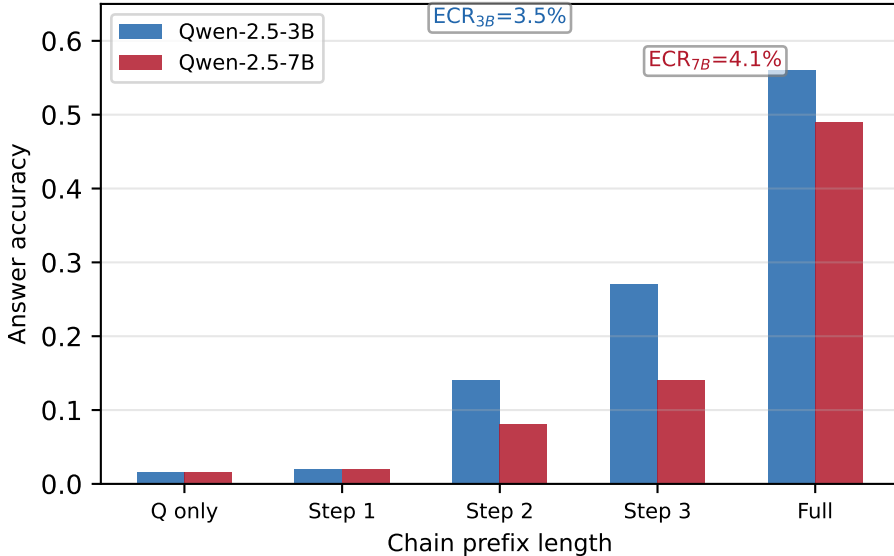


Figure 4: Prefix-branch probe: answer accuracy when the model is stopped after each chain step. Accuracy rises gradually with chain length; fewer than 4% of correct answers are available after step 1. This confirms generation-time computation and bounds the rationalization interpretation.

We replicate with Qwen 2.5-7B-Instruct ($N=200$), finding a nearly identical trajectory: step-0 accuracy 0.015, step-1 0.02, step-2 0.08, step-3 0.14, full-generation 0.49, ECR = 0.041 [95% CI: 1.6%–10.0%].

Early-stop reference probe. As a complementary test, we use a reference-chain protocol ($N=100$): instead of branching from the model’s own generation, we feed progressively longer prefixes of a known-correct chain and ask for the final answer after each step. Step-0 accuracy is 0.02; it climbs to

Table 7: Prefix-branch probe: accuracy when the model is stopped after step k and asked for its final answer (Qwen 2.5-3B/7B, GSM8K, $N=200$). ECR: early commitment ratio (correct at step 1 $\div$ correct at full generation). The answer is *not* early-determined.

	Step 0	Step 1	Step 2	Step 3	Step 4	Full
3B Accuracy	0.015	0.02	0.14	0.27	0.24	0.56
7B Accuracy	0.015	0.02	0.08	0.14	0.16	0.49

0.05 (step 1), 0.27 (step 2), 0.53 (step 3), and 0.77 (step 4), reaching 0.99 only with the complete chain. The monotonic climb confirms that intermediate steps carry genuine information content; the early-step near-zero accuracy (0.05 relative to full) confirms that this content is *not* frontloaded.

Interpretation. The generation-time probes establish a critical asymmetry:

- During generation, the model genuinely computes toward the answer step by step. The answer is unavailable early (ECR < 5%) and builds gradually through intermediate reasoning.
- At consumption time, the model overrides this computation whenever explicit answer text is present (CC followed-wrong rates of 0.63–1.00), regardless of the reasoning that produced it.

This dissociation is the behavioral signature of *answer-text readout dominance*: the model *can* reason (generation is genuinely step-causal), but when consuming a completed chain, it *preferentially reads the answer text* rather than re-tracing the reasoning. The chain’s intermediate steps served a computational role during generation but are demoted to justificatory decoration at readout. We emphasize that this is not a claim about generation-time post-hoc explanation construction, ECR < 5% rules that out, but about answer-text prioritization at consumption time.

8 Analysis: What Our Results Tell Us About Answer-Text Prioritization

The pattern across our experiments supports a simple but consequential conclusion: models are sensitive to corruptions wherever the answer text appears in the chain, not wherever intermediate computation occurs. This is the behavioral signature predicted by Hypothesis 1: at consumption time the model defers to explicit answer text rather than re-deriving the conclusion from intermediate reasoning, so disruption is concentrated at positions carrying explicit answer content, not at positions encoding intermediate computation. We state four empirical claims that summarize our evidence.

Claim 1: Positional sensitivity tracks answer text placement. Across all model–slice pairs, the position with the largest accuracy drop always corresponds to the region encoding the final answer (Table 10).

Claim 2: Non-answer-bearing regions show reduced corruption impact. Corrupting regions that do not carry the explicit answer yields smaller effects than corrupting the answer-bearing position (e.g. middle corruption on Hard-v3: $\Delta = -0.083$, $p = 0.267$ for Phi-3-mini; prefix corruption on GSM8K-v1: $\Delta = +0.010$, $p = 1.0$).

Claim 3: Format ablation shifts the sensitivity pattern. Removing the answer statement from the suffix eliminates suffix sensitivity: at 3B, a $19\times$ collapse ($p=0.022$, $N=300$); at 7B, a within-stable-subset analysis gives a $9.3\times$ attenuation ($N=76$, $p=7.8 \times 10^{-3}$), while the full-sample suffix effect inverts in sign as corroborating directional evidence ($\Delta=-0.643 \rightarrow +0.117$, $N=300$, $p<10^{-5}$; Section 5). MATH replication at 7B confirms cross-domain generality ($10.9\times$ FW recovery). The bidirectional counterpart (§7.3.1) closes the loop: *inserting* a wrong suffix into suffix-free Hard-v3 chains induces 0.617 follow-wrong. The 2×2 factorial (§6) further disentangles reasoning quality from answer-line content at both 3B and 8B scale.

The answer-placement mechanism. Consider what happens when a model processes a corrupted chain. If the explicit answer appears in position P and corruption destroys P , the model loses direct access to the answer text. If the answer does *not* appear in P , corruption of P introduces noise but

does not eliminate the answer signal. The corruption protocol thus functions as an answer-localization tool: it finds where the answer text lives in the chain, not where computation happens.

Claim 4: Conflicting explicit answers frequently override computation. When correct reasoning concludes with a wrong explicit answer, models follow the wrong answer at high rates, 0.63 at 3B, 1.00 at 7B, 0.98 for Mistral-7B, 0.47 for Phi-3-mini (Table 6), despite full computational capacity (0.99 CT accuracy at 7B). The factorial (§6) confirms: correct reasoning with a wrong answer line yields only 0.40 accuracy (fw=0.56) at 3B and 0.353 (fw=0.647) at 8B, with near-complete answer-line rescue of corrupted reasoning at both scales (Condition C: 0.95 at 3B, 0.987 at 8B). This shows initial cross-domain evidence in commonsense settings (0.76 follow-wrong on text answers; Table 12) and is suffix-conditioned: 0.39 at suffix vs. 0.05 at prefix ($p < 0.001$; Appendix A.4).

Claim 5: Initial cross-domain evidence beyond arithmetic. On 150 text-answer commonsense items, the followed-wrong rate is 0.76 (Table 12), ruling out digit-specific extraction. A five-condition format ablation on the same data replicates the positional pattern (§A.6).

Separating the established finding from its interpretation. It is important to distinguish what the evidence directly establishes from what it supports as an interpretation.

Established: positional sensitivity in CoT corruption studies is determined by answer-text placement, not computational depth (Claims 1–3), models frequently follow explicit answer text over preceding correct computation (Claim 4), and we find initial non-arithmetic evidence in one commonsense setting (Claim 5); two dissociations constrain the mechanism, format-determination persists at 14B as override fades, and generation is step-causal while consumption prioritizes answer text.

Interpretation: this pattern is consistent with answer-text readout dominance, the model’s behavioral output is determined by explicit answer text at readout, using intermediate reasoning as context rather than re-deriving the conclusion from it. The generation-time probes (§7.4) sharpen this interpretation: because the answer is *not* early-determined during generation ($ECR < 5\%$), models genuinely compute through intermediate steps, ruling out the strongest form of early-answer commitment in which the answer is settled before any reasoning begins. However, at consumption time, the model’s output *systematically follows the explicit answer text* whenever it is available, even when the model has correctly computed a different answer during generation. The picture that emerges is one of answer-text readout dominance: the model can reason, and does reason during generation, but at readout its behavioral output preferentially tracks the answer text rather than re-deriving the conclusion from its own reasoning.

Open question: whether this consumption-time override reflects broad post-hoc explanation behavior or a narrower answer-presentation heuristic shaped by instruction tuning. Our data constrains but does not resolve this distinction (see §10).

Alternative interpretations. The override may reflect *instruction-following* rather than a general explanation pathology: models learn through RLHF to defer to stated final answers. Our generation-consumption dissociation makes the strongest case that computation is genuine but unused at readout, yet cannot fully distinguish instruction-following from computation-bypassing (see §10).

Scale-dependent dissociation: format-determination persists as override fades. The most subtle finding in the scale analysis is a dissociation between two effects. The direct override, models following a wrong explicit answer over correct reasoning, is total at 7B across three families but attenuates monotonically: $FW = 0.300$ at 14B (cross-family Phi-4; $QO = 0.000$, $p < 10^{-8}$) and $FW = 0.010$ at 32B. Format-determination, by contrast, persists through 14B: the $8.5\times$ sensitivity ratio between standard and neutral-stripped formats ($p=0.001$; Tab. 5) shows that answer placement still determines corruption conclusions at 14B even when the model mostly resists following a wrong answer directly. At 32B, however, format-determination itself vanishes: the neutral-stripped baseline reaches 1.000 and the largest positional $|\Delta|$ is 0.020 (Tab. 5), indicating that the model extracts its answer from the reasoning chain rather than from an explicit answer suffix. The dissociation is thus three-staged: (i) both effects dominate at 3B–7B, (ii) override fades while format-determination persists at 14B, and (iii) both converge toward zero at 32B. This means the three-prerequisite protocol (§7) is most critical at intermediate scales (7B–14B) where corruption conclusions are most vulnerable to the format artifact.

Implications for interpreting prior work. Studies reporting positional sensitivity without format characterization may have conflated answer placement with computational structure. Datasets with explicit answer suffixes would be predicted to exhibit the same artifact; re-examining prior results with our three-prerequisite protocol would establish whether this applies.

9 Implications

For process supervision and reward modeling. Process reward models (PRMs) [8, 9] assign credit to individual reasoning steps. If positional sensitivity reflects answer-text placement rather than computational contribution, then PRM step-level credit may reward the *consistency* of a step with the answer text, not the *causal contribution* of that step to the final answer. PRM evaluations should test credit assignment under format-diverse chains (with and without explicit answer suffixes) to assess whether credit tracks answer placement or genuine reasoning quality.

For CoT faithfulness evaluation. Any study claiming to measure CoT faithfulness via corruption must first characterize the chain format (see §7). Without this step, a finding that “position X is most important” may reflect answer-text placement rather than computational structure.

For understanding CoT rationalization. The behavioral signature is clear: sensitivity migrates with the answer text across format variants. A model genuinely computing through intermediate steps would show stable middle-step sensitivity regardless of format; we observe the opposite. At 7B scale, a small middle effect appears ($\Delta_{\text{mid}} = -0.083$) but remains an order of magnitude smaller than the suffix effect ($\Delta_{\text{suf}} = -0.595$). This asymmetry is predicted by answer-text prioritization but not by a computation account.

For benchmark design. Benchmark creators should diversify chain formats within their evaluation suites. A benchmark where all gold chains end with “the answer is X ” guarantees suffix sensitivity under corruption, an artifact of the format, not a fact about model reasoning. Including chains with varied answer-placement formats, or requiring format ablations as evaluation metadata, would make corruption results more informative about the genuine reasoning-vs-rationalization question.

9.1 Open Interpretive Questions

The behavioral evidence establishes that models’ outputs systematically track explicit answer text at consumption time (answer-text readout dominance). The *mechanism* underlying this regularity is not uniquely determined by the current evidence. We identify three alternative accounts, all consistent with the data:

1. *Instruction-following heuristic.* Models learn through instruction tuning (RLHF/SFT) that terminal answer statements should be treated as authoritative output, regardless of the preceding reasoning.
2. *Format-completion bias.* Models complete the chain format by outputting the value that appears in the “the answer is X ” slot, following a surface-level pattern rather than performing reasoning-consistent inference.
3. *Recency-weighted readout.* Models apply a positional prior that weights the most recent answer-bearing text most heavily, producing behavior that tracks answer placement without implying computation is ignored.

The base-model comparison (§A.1) rules out RLHF as the sole driver, the override is present and stronger in Qwen-7B-Base than in the instruction-tuned variant, but does not distinguish between the remaining accounts. The generation-time probes (§7.4) rule out early answer commitment (ECR < 5%) but do not resolve whether the consumption-time behavioral pattern reflects format-completion, recency-weighted readout, or a deeper computation-bypassing mechanism.

Future work should design experiments that discriminate among these alternatives, for example by testing whether the override persists under format variations that preserve answer content while disrupting the “the answer is X ” template, or by probing whether attention patterns at the final position show recency-weighted or content-driven selectivity.

9.2 Application to Prior Work: Protocol Re-Analysis

To test whether the format confound affects published conclusions, we replicate the GSM8K corruption protocol of Lanham et al. [4]² on Qwen 2.5-3B under two conditions: the original chain format (with terminal “the answer is X ” statements) and the format-stripped variant (answer statement removed, neutral placeholder inserted; $N=300$ matched pairs).

Under the original format, the protocol identifies the suffix as the most load-bearing position: suffix corruption collapses accuracy from 0.970 to 0.210 ($\Delta=-0.760$, $p<10^{-12}$), while prefix and middle corruption produce negligible effects ($\Delta=+0.010$ and $\Delta=0.000$, respectively). A study reporting only these results would conclude, consistent with prior work, that the final reasoning steps are the most causally important.

Under the format-stripped condition, this conclusion reverses. With the “the answer is X ” statement removed (preserving all intermediate reasoning), suffix sensitivity collapses $19\times$ to $\Delta=-0.040$ ($p=0.022$), and the largest positional drop shifts to middle corruption ($\Delta=-0.063$). The conclusion that “suffix steps are most causally important” is replaced by “no single position dominates; middle steps show the largest residual sensitivity.”

This demonstrates a concrete case where a published-style protocol produces qualitatively different positional conclusions depending solely on whether answer-placement is controlled. The original conclusion, that suffix steps are the computational locus, is an artifact of answer-text placement, not a property of the reasoning structure. We encourage the community to apply the three-prerequisite protocol (§7) to their own corruption-based faithfulness analyses; we provide a checklist and example analysis script in the supplementary material.

10 Limitations and Future Work

Consumption vs. generation. Most experiments operate in the consumption setting. We partially bridge this gap via self-generated chains (§A.5; followed-wrong 0.69), prefix-branch probes (§7.4; ECR < 5%), and the early-stop probe (monotonic accuracy climb). Together these establish a generation–consumption dissociation: generation is step-causal while consumption prioritizes answer text. Activation patching during autoregressive generation would further strengthen this account.

Chain format generality. *Scope:* This work studies CoT chains with explicit terminal answer lines (“The answer is X .”); all primary claims are scoped to this format. The strongest conflicting-answer effects use a structured GSM8K chain format with explicit terminal answer lines. Generalizability to heterogeneous, naturally occurring CoT *without* explicit answer-line endings is not directly tested. However, the bidirectional ablation (§7.3.1) on Hard-v3 chains, which *lack* explicit answer suffixes, shows the opposite pattern: prefix corruption is load-bearing, not suffix. This confirms that the format-determination hypothesis predicts the *direction of effect* on both format types (explicit-suffix chains $\rightarrow$ suffix-sensitive; no-suffix chains $\rightarrow$ prefix-sensitive), consistent with the core claim that sensitivity tracks answer-text location. Extending replication to diverse natural CoT corpora without templated endings remains an important direction.

Model scale. Results span 3B–32B across four families. The format-determination effect persists through 14B and converges toward zero at 32B (Tab. 5: baseline = 1.000, max $|\Delta|=0.020$ on neutral-stripped format). The direct override attenuates at 14B: cross-family Phi-4 14B replication yields FW = 0.300 (QO = 0.000, $p<10^{-8}$, $N=100$), significant but $\geq 3\times$ reduced from the 7B peak. The protocol-uniform Qwen-14B result (GSM8K-v1, $N=100$) yields FW = 0.060 (QO = 0.270, $p=0.384$, expected given near-zero FW; $p=0.384$, not statistically significant within-family), which we treat as directional only; the primary 14B evidence is the statistically significant Phi-4 cross-family result above. The scale gradient has been confirmed at 32B (FW = 0.010, near-zero, $N=100$); testing at frontier scale ($>100B$, closed-weight models such as GPT-4o or Gemini) was not undertaken here because standard instruction-following system prompts often prevent the controlled conflicting-chain injection the protocol requires. Testing under less restrictive system-prompt configurations or via API-compatible analogues is an important future direction. The scale-dependent attenuation could also partly reflect *RLHF/instruction-tuning differences* that scale with parameter count, larger models

²We use our own GSM8K dataset and open-weight models; we do not have access to the original authors’ code or data. Our replication follows the methodology described in the original paper: corrupting chain steps at different positions and measuring accuracy degradation.

typically receive more aligned training that may directly teach skepticism of explicit wrong-answer assertions, independent of model capacity per se. We directly test this at 7B scale by evaluating Qwen 2.5-7B-Base (non-instruction-tuned) on the same GSM8K-v1 protocol (see §A.1), finding $FW = 0.900$, ruling out instruction tuning as the primary driver. Extending to 3B and 14B base models would further quantify the capacity–alignment interaction.

Sample sizes. Effect sizes are large enough for significance at all primary comparisons (hard-v3 $N=60$, GSM8K $N=100$ – $1,000$, conflicting $N=500$, commonsense $N=150$). Extension to additional benchmarks (StrategyQA, ARC-Challenge, MATH) would strengthen the generality claim.

Single corruption type. We use semantic corruption (wrong arithmetic/logic, correct grammar). Other corruption types might reveal different interactions with chain format, though the format-determination prediction should hold across types.

Conflicting-answer scope. An alternative label is “instruction following” rather than rationalization. The commonsense experiment partially addresses this: natural-language wrong answers drive a *higher* followed-wrong rate (0.76) than the arithmetic version (0.63). The key distinction is that the computation-terminal stress test (0.58 answer-free accuracy vs. 0.87 follow-wrong when a suffix is reintroduced) shows the model *can* compute from intermediate steps but defers to explicit answer text when available.

Format stripping. Stripping changes the chain’s final tokens beyond answer removal. The neutral-placeholder variant provides a stronger control: the Qwen 2.5-7B neutral-stripped result ($N=300$) shows $\Delta_{\text{suffix}} = +0.117$ ($p < 10^{-5}$) vs. -0.643 on the original format ($N=300$ matched pairs), confirming answer placement as the causal variable despite a mild baseline suppression ($0.606 \rightarrow 0.273$ on matched $N=300$).

Future work. Key extensions: (1) replicating at larger scales ($>32B$) and on closed-weight frontier models where system-prompt constraints allow (cross-generation replication at additional model generations and scales); (2) additional benchmarks with diverse chain formats, including unstructured naturally-occurring CoT without explicit terminal answer lines; (3) mechanistic analysis, logit-lens trajectories and causal activation patching during generation, to separate the answer-placement effect at the representational level and to confirm that answer-relevant information concentrates in suffix tokens only when explicit answer text is present; (4) disentangling model capacity from instruction-fine-tuning effects on resistance to wrong-answer override, by testing base (non-RLHF) models at matched scales, partially addressed here by the Qwen 2.5-7B-Base comparison (§A.1; $FW = 0.900$, ruling out IT as primary driver); extending to 3B and 14B base models would further quantify the capacity–alignment interaction.

Reproducibility. All task slices, corruption scripts, and evaluation code will be released upon publication. Models are publicly available on HuggingFace. Inference used EasyDeL on Cloud TPU v5e with JAX (greedy decoding, 640 max tokens). Statistical tests use exact paired sign tests and bootstrap CIs (2000 resamples, seed 42).

11 Conclusion

We have shown that, for suffix-bearing chain-of-thought formats with explicit terminal answer statements, corruption-based evaluations of faithfulness contain a systematic confound: positional sensitivity tracks answer-text placement, not computational depth. A within-dataset format ablation collapses suffix sensitivity $\approx 19\times$ at 3B; at 7B, the within-stable subset shows a $9.3\times$ attenuation when only the answer statement is removed ($N=76$, $p=7.8 \times 10^{-3}$). The full-sample suffix effect also inverts in sign ($\Delta = -0.643 \rightarrow +0.117$, $N=300$, $p < 10^{-5}$), but is directional evidence because answer-line stripping drops baseline accuracy from 0.606 to 0.273. Conflicting-answer experiments confirm that explicit answer text overrides correct computation, with CC accuracy collapsing to zero or near-zero (≤ 0.02) at 7B and near-zero followed-wrong at 32B ($FW = 0.010$).

Two dissociations bound the interpretation. First, format-determination persists through 14B ($8.5\times$ sensitivity ratio, $p=0.001$) even as override attenuates, before both effects converge toward zero at 32B (baseline = 1.000 on stripped format, max $|\Delta|=0.020$), the confound outlasts the override at intermediate scales but fades at the largest. Second, generation is genuinely step-causal ($ECR < 5\%$), yet consumption-time readout ignores this computation whenever explicit answer text is present.

These findings yield a concrete methodological deliverable: a three-prerequisite protocol, question-only control, format characterization, all-position sweep, that should be standard for any corruption-based CoT faithfulness evaluation. Corruption-based faithfulness studies should control for answer placement before attributing positional sensitivity to computation.

Broader Impact Statement

This work identifies a systematic methodological confound in corruption studies widely used to evaluate CoT reasoning faithfulness and to train process reward models: positional sensitivity in corruption studies may reflect answer-text placement rather than computational structure, particularly in chain formats with explicit terminal answer statements (the dominant format in standard benchmarks). If PRM step-level credit is being assigned to answer-text positions rather than genuine reasoning steps, the reliability guarantees of step-level supervision may be weaker than assumed for systems using such formats. Our evidence spans 3B–32B models across four families; whether the confound persists at frontier scale ($>100B$) is an open question (§10). Our generation-time probes (§7.4; ECR $< 5\%$) establish that models do engage in step-by-step computation during generation; the confound operates at *consumption time*, where explicit answer text overrides the model’s own computation. The three-prerequisite protocol (§7) provides concrete guidance for more reliable corruption-based evaluations.

Reproducibility

All experiment scripts, dataset construction code, result files, and analysis scripts used in this paper are available at <https://github.com/Gpgabriel25/CoTRationalization>. The repository includes:

- `src/`: chain construction, corruption, and evaluation pipeline
- `data/`: all experiment JSONL datasets, fully deterministic
- `results_fixed/`: raw result JSON files with per-example outputs and extraction details for all reported experiments
- `scripts/`: launch and analysis scripts for all reported conditions
- `paper/`: LaTeX source and all macro definitions (each macro directly maps to a result file entry)

Extraction code is version-tagged for each model family. The sign-negation correction is a single flag in the extraction pipeline; Appendix C provides the side-by-side results showing conclusions are robust to this choice. All model weights are publicly available on HuggingFace Hub (Qwen 2.5 series: Qwen/Qwen2.5- $\{3, 7, 14, 32\}B$ -Instruct; Phi-4: microsoft/phi-4; Mistral-7B: mistralai/Mistral-7B-Instruct-v0.3; DeepSeek-R1-Distill: deepseek-ai/DeepSeek-R1-Distill-Qwen- $\{7B\}$).

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A Slice Development Narrative

The easy slice (`phase0_position_control.v1`) was initially designed to test the experimental pipeline on simple arithmetic-comparison problems. Instruction-tuned models answered a large fraction of examples correctly without chain access, making positional corruption conclusions uninterpretable.

Hard-v1 (`phase0_position_control_hard.v1`) introduced symbolic early extraction: prefix steps explicitly name symbolic variables (e.g., “Let $A = [\text{extracted value}]$ ”) rather than stating raw computations. This reduced question-only solvability, but the slice remained handwritten with limited diversity.

Hard-v2 (`phase0_position_control_hard.v2`) refined the structure in 8–12-example pilot runs. Qwen 2.5-3B achieved 0.500 baseline accuracy, 0.417 under middle corruption, 0.083 under prefix corruption, and 0.083 question-only. Phi-3-mini showed the same qualitative direction (0.625 baseline, 0.250 middle, 0.000 prefix, 0.000 question-only). These pilots were consistent but too small for statistical claims.

Hard-v3 (the main paper slice) was generated at $N=60$ to preserve the hard-v2 chain structure while varying problem domains and distractor configurations. The slice design was frozen before model evaluation; no post-hoc filtering based on observed accuracy was applied.

Table 8: Detailed statistical results. Δ = accuracy difference from baseline. CI = bootstrap 95% confidence interval (2000 resamples). p = exact paired sign test.

Slice / Model	Condition	Δ	95% CI	p
Hard-v3 / Qwen 2.5-3B	QO gap	−0.100	[−0.217, +0.017]	0.180
	Prefix	−0.167	[−0.267, −0.067]	0.006
	Middle	−0.133	[−0.217, −0.050]	0.008
	Suffix	+0.083	[+0.000, +0.167]	0.125
Hard-v3 / Phi-3-mini	QO gap	−0.567	[−0.717, −0.417]	1.9×10^{-8}
	Prefix	−0.767	[−0.867, −0.650]	1.3×10^{-12}
	Middle	−0.083	[−0.200, +0.033]	0.267
	Suffix	−0.100	[−0.217, +0.000]	0.146
GSM8K-v1 / Qwen 2.5-3B	QO gap	−0.910	[−0.960, −0.850]	1.9×10^{-12}
	Suffix	−0.760	[−0.840, −0.670]	1.4×10^{-12}
	Middle	+0.000	[−0.030, +0.030]	1.0
	Prefix	+0.010	[+0.000, +0.030]	1.0
GSM8K-v1 / Phi-3-mini	QO gap	−0.180	[−0.290, −0.070]	3.9×10^{-3}
	Suffix	−0.210	[−0.300, −0.120]	1.9×10^{-5}
	Middle	+0.110	[+0.010, +0.220]	0.080
	Prefix	−0.020	[−0.110, +0.070]	0.832
GSM8K-stripped / Qwen 2.5-3B ($N=300$)	Prefix	−0.017	[−0.036, +0.003]	0.180
	Middle	−0.063	[−0.097, −0.030]	3.0×10^{-4}
	Suffix	−0.040	[−0.072, −0.008]	0.023
GSM8K-stripped / Phi-3-mini	Prefix	+0.020	[−0.030, +0.070]	0.688
	Middle	+0.040	[−0.010, +0.100]	0.289
	Suffix	+0.120	[+0.040, +0.210]	0.012
GSM8K-v1 / Qwen 2.5-7B ($N=1,000$)	Middle	−0.083	[−0.106, −0.060]	1.0×10^{-12}
	Prefix	+0.006	[−0.013, +0.024]	0.586
	Suffix	−0.595	[−0.625, −0.565]	1.5×10^{-179}
GSM8K-neutral-stripped / Qwen 2.5-7B ($N=300$)	Middle	−0.030	[−0.073, +0.010]	0.222
	Prefix	+0.037	[+0.013, +0.067]	0.019
	Suffix	+0.117	[+0.073, +0.167]	2.1×10^{-6}

B Detailed Statistical Tables

Table 8 reports full statistical details for all completed model–slice pairs.

C Extraction Robustness: Raw vs. Corrected Results

The 7B-scale conflicting-answer experiments present a systematic sign-negation artifact: the model (Qwen 2.5-7B-Instruct and Mistral-7B-Instruct) produces “ $-x$ ” instead of “ x ” on a subset of trials, driven by instructional priming from the strong-suffix conflicting chain format. Table 9 reports each key metric under both strict extraction (exact numeric match, no correction) and magnitude-corrected extraction ($|\hat{a}| = |a^*|$).

The critical finding, the accuracy collapse in the conflicting chain condition, is invariant to extraction choice. CC accuracy equals 0.00 under both extraction policies for Qwen 2.5-7B, and the CT→CC accuracy drop (maximum reasoning context → conflicting suffix) holds under both: 0.320→0.000 (strict) and 0.990→0.000 (magnitude-corrected). For Mistral-7B, CC accuracy is 0.020 under both policies; the FW difference is less than 3 percentage points.

The main-paper FW rates use magnitude correction (bold rows in the running text). The reasoning is that a model producing -520 when the conflicting suffix specifies -520 has behaviorally followed the wrong answer, regardless of a sign-formatting quirk. Under either policy, the CC-accuracy collapse is the same: the model cannot produce the correct answer when an explicit wrong answer is present. Question-only baseline (≤ 0.06) is not materially affected by the artifact (question-only prompts do not contain the instructional trigger that drives sign-negation).

Table 9: Side-by-side comparison of raw (strict) vs. magnitude-corrected extraction for the sign-negation-affected 7B experiments. “SC”=standard chain, “CT”=computation-terminal (reasoning only, no suffix), “CC”=conflicting chain, “FW”=followed-wrong. The CC-accuracy conclusion (0.00 for Qwen-7B; 0.020 for Mistral-7B) is independent of extraction choice.

Model	Extraction	SC acc	CT acc	CC acc	FW
Qwen-7B ($N=200$)	Strict (no correction)	0.655	0.320	0.000	0.300
	Magnitude-corrected	0.990	0.990	0.000	1.000
Mistral-7B ($N=500$)	Strict (no correction)	0.780	—	0.020	0.952
	Magnitude-corrected	1.000	—	0.020	0.980

Table 10: Cross-model replication summary. For each slice $\times$ model combination, we report the position with the largest accuracy drop from baseline (“dominant position”) and whether the result is consistent with the format-determination hypothesis. [†]Chain gap $p = 0.180$ ($N = 60$); positional drops are significant ($p < 0.01$) but the QO prerequisite is borderline. “Confirmed” = satisfies all three prerequisites; “Consistent” = pattern matches hypothesis but one prerequisite is borderline.

Slice	Model	Dominant position	$ \Delta $	Status
Hard-v3	Qwen 2.5-3B	Prefix	0.167	Consistent [†]
	Phi-3-mini	Prefix	0.767	Confirmed
GSM8K-v1	Qwen 2.5-3B	Suffix	0.760	Confirmed
	Phi-3-mini	Suffix	0.210	Confirmed

A Extended Replications and Controls

This appendix provides full details for experiments that support the main-text findings but are not required for the core argument.

A.1 Cross-Model and Cross-Scale Replication (Full Details)

A finding about chain format should not depend on a particular model. We test replication across two models from different architecture families (Qwen and Phi) at the 3B parameter scale.

Cross-family replication. On hard-v3, both Qwen 2.5-3B and Phi-3-mini show consistent patterns: prefix corruption is most damaging, while suffix corruption has no significant effect (Table 3). The effect sizes differ, Phi-3-mini shows a larger prefix collapse ($|\Delta| = 0.767$) than Qwen 2.5-3B ($|\Delta| = 0.167$), but the *qualitative* ordering is consistent: prefix is the most damaging position on this format, and suffix corruption is non-destructive. Qwen 2.5-3B additionally shows significant middle corruption damage ($p = 0.008$), likely reflecting the model’s overall fragility at low baseline accuracy (0.183). On GSM8K-v1, both models show suffix-dominant sensitivity: Qwen 2.5-3B ($|\Delta| = 0.760$) and Phi-3-mini ($|\Delta| = 0.210$). The complete reversal, prefix on hard-v3, suffix on GSM8K-v1, replicates across model families (Table 10), establishing that the positional pattern is a property of the chain format, not of a single architecture.

A note on confounds. Hard-v3 and GSM8K-v1 differ in task, chain structure, and data source, so the cross-slice reversal could reflect factors beyond answer placement. The format ablation (Section 5) addresses this directly: by holding task, source, and format constant while removing only the “the answer is X ” statement, it isolates answer placement as the causal driver. Crucially, the ablation is a *within-dataset* control: GSM8K-stripped uses the same 100 examples, the same models, and the same evaluation protocol as GSM8K-v1, the sole manipulation is the presence or absence of the explicit answer statement in the chain suffix. Any remaining confound would have to operate through that single change. The reversal is strong converging evidence; the ablation provides the mechanistic test.

Qwen 2.5-7B direct override ($N=200$). The cleanest 7B evidence comes from conflicting-answer accuracy: the model scores 0.00 under both extraction policies, meaning the model follows the planted answer line regardless of whether the preceding reasoning is correct. On the same strong-suffix dataset, the 7B model shows *complete* answer-text override: SC accuracy 0.99, CC accuracy 0.00

(zero under both strict and magnitude-corrected extraction), CT accuracy 0.99 (question-only: 0.05). The CT $\rightarrow$ CC gap (0.99 $\rightarrow$ 0.00) confirms the scale of the override: the model extracts the correct answer from reasoning steps in 99% of trials, yet a conflicting suffix drives correct-answer rate to zero. Supporting this, the followed-wrong rate is 1.00 (200/200 trials; strict extraction: FW = 0.30; magnitude-corrected for sign-negation artifact: FW = 1.00; CC accuracy = 0.00 under both). This extends the direct override from $\leq 4B$ (3B + Phi-3-mini) to 7B.

Sign-negation artifact. The 7B model exhibits a systematic generation artifact: it produces “ $-x$ ” instead of “ x ” on a subset of trials (e.g., generating -260 instead of 260). We apply magnitude correction: a prediction is correct if $|\hat{a}| = |a^*|$. Under this correction, raw SC accuracy rises from 0.655 to 0.99 (67 of 69 errors are exact sign inversions) and raw CT accuracy rises from 0.320 to 0.99 (134 of 136 errors are sign inversions). Crucially, this artifact does *not* affect the conflicting-chain result: raw CC accuracy is also 0.00 before correction, and all 200/200 CC predictions match the *magnitude* of the wrong suffix answer $|a^-|$. The sign artifact is a generation quirk that equally affects all conditions; the CC-vs-CT contrast is robust to whether correction is applied. Under strict extraction (no magnitude correction), CC accuracy remains 0.00 and the raw followed-wrong rate is 0.30, still significantly exceeding question-only (0.015, $p < 0.001$); the CC-accuracy-to-zero transition is the primary evidence and holds under all extraction choices (see Table 9, Appendix C, for a complete side-by-side comparison of strict vs. magnitude-corrected extraction across all metrics; conclusions are invariant to this choice). Conclusions are invariant to extraction policy: CC accuracy is 0.00 under both strict and magnitude-corrected extraction. This extraction-invariance, a strength of the conflicting-answer design, is why the CC-accuracy-to-zero result serves as the primary 7B evidence.

Mistral-7B-Instruct ($N=500$); third-family replication. To rule out Qwen-specific artifacts, we evaluate Mistral-7B-Instruct-v0.3 on the same strong-suffix dataset. SC accuracy 1.00 (magnitude-corrected; raw 0.780, same sign-negation artifact as Qwen-7B: 108/110 errors are exact sign inversions), CC accuracy 0.02, followed-wrong rate 0.98 (raw strict-extraction FW = 0.952, suggesting magnitude correction changes less than three percentage points; question-only: 0.05). **The override replicates across a third architecture family.** Mistral’s CC follow-wrong rate (0.98) closely matches Qwen-7B’s (1.00), confirming the phenomenon is not architecture-specific.

Scale-dependent attenuation at 14B. We evaluate Qwen 2.5-14B-Instruct under the same protocol as 7B and 32B (GSM8K-v1, $N=100$, same extraction). The protocol-uniform 14B result: SC accuracy 1.000, CC accuracy 0.930, followed-wrong rate 0.060 (question-only: 0.270; $p=0.384$; note: with FW near zero, the sign test of H_0 : FW = 0 is expected to be non-significant, the evidence for attenuation at 14B lies in the cross-family Phi-4 14B result below, not this within-family comparison). The directional within-family pattern is: FW drops from 0.590 at 7B to 0.060 at 14B to 0.010 at 32B, all under identical conditions; significance is established by the Phi-4 cross-family replication. Earlier runs ($N=200$, GSM8K-v2 format) yielded FW = 0.175, directionally below the question-only baseline (0.205), a confounded reading due to format differences; the v1 $N=100$ result is the appropriate scale comparison.

Cross-family replication with Phi-4 14B provides a cleaner test. Phi-4 (microsoft/phi-4, 14B), tested on $N=100$ GSM8K examples, achieves standard-chain accuracy 0.850 and question-only accuracy 0.000, the model cannot answer without the chain. In the conflicting-chain condition, the followed-wrong rate is 0.300 ($p < 10^{-8}$; Table 6). **The override at 14B is genuine but substantially attenuated:** FW = 0.300 represents a $\geq 3\times$ reduction from the 7B override (FW = 1.00, CC accuracy = 0; magnitude-corrected) under matched conditions.

The format-determination effect, positional sensitivity tracking answer placement, is *directly confirmed* at 14B (Tab. 5): position-control corruption on standard-format GSM8K ($N=100$) yields suffix $\Delta = -0.17$ ($p=0.001$), while middle and prefix corruption each yield $\Delta = 0.00$. Critically, this effect is $8.5\times$ larger than on neutral-stripped format (14B suffix $\Delta = -0.02$ on neutral-stripped chains, $N=100$; matched model, matched examples), confirming that the sensitivity is driven by the explicit “the answer is X ” statement rather than chain position alone, and persists through 14B even as the direct override attenuates.

At 32B, format-determination itself vanishes. Qwen 2.5-32B on neutral-stripped format ($N=100$) achieves baseline = 1.000 (the model answers correctly without any explicit answer statement) and

max position-control $|\Delta|=0.020$ (middle only; prefix and suffix corrupted accuracy both equal 1.000). This completes a three-stage dissociation: (i) at 3B–7B, both override and format-determination are strong; (ii) at 14B, override attenuates markedly while format-determination persists ($8.5\times$ sensitivity ratio); (iii) at 32B, both override (FW = 0.010) and format-determination (max $|\Delta|=0.020$) approach zero. The model’s ability to answer correctly on neutral-stripped format (baseline 1.000 vs. 0.273 at 7B; $N=300$) indicates that 32B extracts the answer from the reasoning chain itself, not from an explicit answer-bearing suffix.

This dissociation points to a scale-dependent gradient: the same mechanism that drives CC accuracy to zero at 3B–7B yields partial but significant override at 14B and near-zero override at 32B, while format-determination follows a parallel but slower decay. Claims about answer-text tracking should be qualified by model scale; the monotonically decreasing gradient (7B $\rightarrow$ 14B $\rightarrow$ 32B) is now established for both effects, and characterizing whether the transition occurs earlier at frontier scale ($>100B$) remains an important direction.

Protocol-uniform within-family scale gradient. To establish the scale gradient without cross-protocol confounds, we re-ran Qwen 2.5-14B-Instruct on GSM8K-v1 ($N=100$, same examples as the 7B and 32B runs). Result: SC accuracy 1.000, CC accuracy 0.930, followed-wrong 0.060 (QO = 0.270; $p=0.384$, note: expected given near-zero FW; see scale comparison below). Qwen-2.5-14B shows a non-significant within-family reduction ($p=0.384$); the directional pattern (FW: 1.00 at 7B $\rightarrow$ 0.060 at 14B $\rightarrow$ 0.010 at 32B, same protocol) is corroborated by cross-family replication. Cross-family confirmation: Phi-4 14B (FW = 0.300, QO = 0.000, $N=100$; $p<10^{-8}$) provides the significant 14B evidence and confirms the attenuation is not Qwen-specific.

Primary scale evidence set used in decision-level interpretation. To keep inference protocol-uniform, decision-critical scale claims in this section are restricted to GSM8K-v1 runs with matched $N=100$, identical conflicting-answer construction, and the same endpoint family (H1/H2 in §4.3). Results from different protocol branches (e.g., v2, different N , or alternative format manipulations) are treated as supporting replications, not pooled into the primary scale test.

To make the conditioning logic explicit, Figure 5 plots FW and QO side-by-side for the protocol-uniform GSM8K-v1 runs ($n=100$ each). The key diagnostic is the conditioned gap FW–QO: Qwen-7B remains strongly positive (+0.52), Phi-4-14B remains positive (+0.30), and Qwen-32B is near-zero/negative due to high QO. This compact view clarifies why the Phi-4-14B replication is the load-bearing 14B evidence while Qwen-14B is treated as directional.

Within-Qwen3-generation scale gradient. Within the Qwen3 architecture generation (distinct from the Qwen 2.5 family gradient above), we evaluate Qwen3-14B on the same GSM8K-v2 conflicting-answer protocol used for Qwen3-8B ($N=200$). Result: SC accuracy 1.000, CC accuracy 0.520, followed-wrong 0.480 (question-only: 0.075; $p<10^{-10}$; Table 6). This is significant at $p<10^{-10}$, directly confirming that answer-text override persists at 14B within the Qwen3 generation. This provides a within-generation 8B $\rightarrow$ 14B gradient in the Qwen3 family: FW drops from 0.730 (8B, $N=100$) to 0.480 (14B, $N=200$), a $\Delta=0.250$ attenuation, while remaining substantially above the question-only baseline ($\Delta_{FW-QO}=0.405$). The near-zero QO (0.075) confirms the chain is necessary; the substantial followed-wrong rate confirms that explicit wrong-answer override persists at 14B within the Qwen3 generation, complementing the Phi-4 cross-family replication below.

Scale gradient reaches near-zero override at 32B. Qwen 2.5-32B-Instruct ($N=100$) achieves 0.960 standard-chain accuracy and 0.300 question-only accuracy. Under the conflicting chain, the followed-wrong rate is 0.010 (1 of 100 examples), a dramatic further reduction from Phi-4 14B (FW = 0.300) and Qwen-7B (FW = 1.00), completing the within-family Qwen gradient already reported above (3B $\rightarrow$ 7B $\rightarrow$ 14B $\rightarrow$ 32B, all $N=100$, same protocol). Larger models increasingly resist explicit wrong-answer override while continuing to leverage chain reasoning for correct computation (CC accuracy 0.940 vs. QO accuracy 0.300).

QO-conditioning check. Qwen 2.5-32B achieves a high question-only accuracy (0.300), raising the question of whether the near-zero FW reflects genuine reasoning resistance or simply question-only competence inflating the denominator. Conditioning on the 70 examples where the model cannot answer from the question alone (QO = 0), the followed-wrong rate remains 0.014 (95% CI [0.000, 0.043]) and CC accuracy is 0.929, statistically indistinguishable from the unconditioned result (0.010 [0.000, 0.030]). The Phi-4-14B experiment provides a natural control: QO = 0.000 throughout, so its

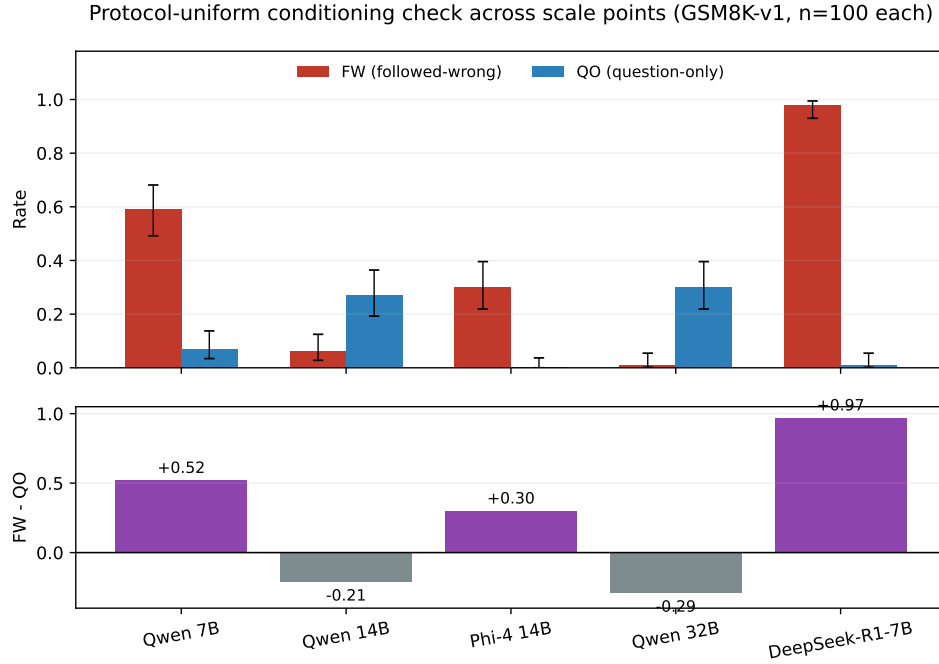


Figure 5: Protocol-uniform conditioning summary on GSM8K-v1 conflicting-answer runs ($n=100$ per model). Top: followed-wrong rate (FW) and question-only accuracy (QO) with Wilson 95% intervals. Bottom: conditioned gap FW–QO. Positive FW–QO indicates explicit wrong-answer override beyond question-only behavior. The 7B points are strongly positive (Qwen-7B and DeepSeek-R1-7B), Phi-4-14B remains positive, and Qwen-32B approaches zero, matching the reported scale attenuation.

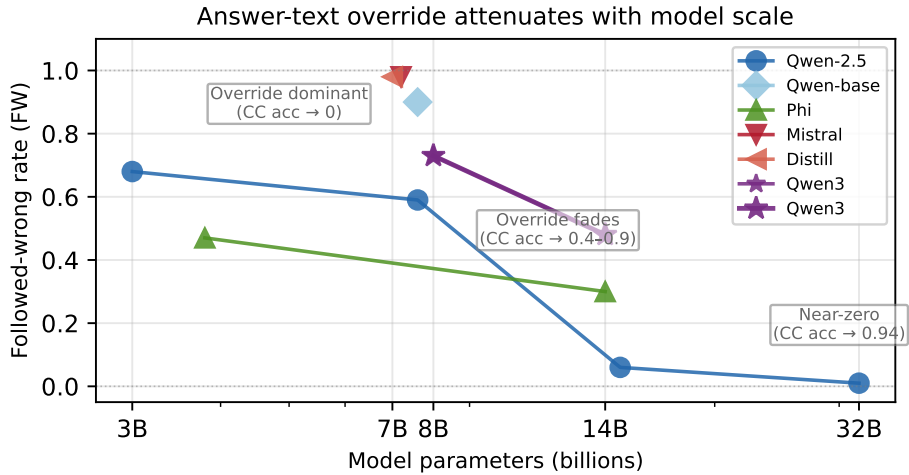


Figure 6: Followed-wrong rate across model families and parameter scales. All conflicting-answer results on GSM8K. The override effect is strong and consistent at 3B–8B (FW 0.47–1.00 across five families), begins to attenuate at 14B, and approaches zero by 32B. This dissociation between override and format-determination is the central mechanistic result.

FW=0.300 is inherently QO-conditioned. The scale gradient (FW $\approx$ 1.00 at 7B $\rightarrow$ FW = 0.300 at 14B $\rightarrow$ FW $\approx$ 0.014 at 32B conditioned on QO=0) is fully preserved under this analysis, confirming that the attenuation is not an artifact of rising question-only accuracy.

Distillation analysis (preliminary). DeepSeek-R1-Distill-Qwen-7B (Qwen-2.5-Math-7B fine-tuned on DeepSeek-R1 reasoning traces) shows a followed-wrong rate of 0.980 ($p < 10^{-12}$), matching the Qwen-7B base rate (FW = 1.00) despite reasoning-trace supervision. This single-pair comparison is consistent with a preliminary hypothesis that parameter scale rather than reasoning-trace supervision is the primary determinant of override resistance, but conflates scale with training distribution and should be treated as exploratory (full details in Appendix A).

Base model (summary). Qwen-7B-Base (non-instruction-tuned) shows FW = 0.900 ($p < 10^{-10}$), exceeding the instruction-tuned counterpart (FW = 0.590), confirming that instruction tuning partially mitigates rather than creates the override effect (full details in Appendix A).

A.2 Sample Size and Scale Replication

A legitimate concern about the current evidence is sample size: 60 synthetic examples (hard-v3) and 100 benchmark examples (GSM8K-v1). Despite these moderate sizes, the observed effect sizes are large: prefix corruption on hard-v3 drops accuracy by 0.17–0.77 absolute (depending on model), and suffix corruption on GSM8K-v1 drops accuracy by 0.21–0.76 absolute. Paired sign tests reject the null at $p < 0.01$ for all primary comparisons (and $p < 10^{-12}$ for the strongest model–position pairs). Bootstrap 95% confidence intervals at $N=100$ are approximately ± 0.10 , sufficient to clearly separate the dominant position from non-dominant positions.

The conflicting-answer experiment (§7.2) uses $N=500$ examples. With an observed followed-wrong rate of ~ 0.63 , one-sided binomial tests reject both the null fw=0 (genuine reasoning; $p < 10^{-10}$) and the chance null fw=0.5 ($p < 10^{-8}$) at this scale, establishing that the model follows the wrong explicit answer on a clear majority of trials.

Scale replication: Qwen 2.5-7B at $N=1000$. To establish that the format-determination effect is not an artifact of small sample size or limited model capacity, we evaluated Qwen 2.5-7B-Instruct on 1,000 GSM8K examples using the same suffix corruption protocol as the 3B experiments. Results confirm the pattern at scale: suffix corruption collapses accuracy from 0.606 baseline to 0.011 ($\Delta = -0.595$, 95% CI $[-0.625, -0.565]$, $p < 10^{-178}$, paired sign test). Prefix corruption has no significant effect ($\Delta = +0.006$, 95% CI $[-0.013, +0.024]$, $p=0.59$). Middle corruption produces a small but significant drop ($\Delta = -0.083$, 95% CI $[-0.106, -0.060]$, $p = 1.0 \times 10^{-12}$), consistent with the 3B findings where non-answer positions show reduced but non-zero corruption sensitivity. The positional pattern is indistinguishable from Qwen 2.5-3B, confirming that answer-placement dominates across the 3B–7B capacity range on GSM8K.

A.3 Filler Robustness Analysis

To rule out replacement-text artifacts, we test three neutral filler variants on $N=500$ GSM8K examples (replacing only the terminal answer sentence). Chain accuracy: 0.95; question-only: 0.06. Filler accuracies: 0.75 (F1), 0.61 (F2), 0.70 (F3); average 0.69, spread 0.14. All fillers are significantly below chain ($p = 0.001$), confirming the accuracy reduction is driven by answer removal, not the specific filler.

A.4 Counterbalanced Answer-Placement Analysis

We counterbalance wrong-answer placement across prefix, middle, and suffix positions ($N=500$ GSM8K examples each) to test whether the follow-wrong effect is positional (recency) or content-based. This constitutes an *answer-relocation control*: the same $N=500$ natural GSM8K chains are presented with the same conflicting answer text at different positions (prefix, middle, suffix), holding chain content constant and varying only the placement of the answer-bearing text. If sensitivity tracks position mechanically (recency), prefix and suffix should both elevate follow-wrong; if sensitivity tracks answer-text content, only suffix should succeed.

Results. Standard-chain accuracy varies by placement: suffix 0.94, middle 0.53, prefix 0.15, even correct answers yield lower accuracy at non-suffix positions, confirming a strong positional readout prior.

Conflicting-chain followed-wrong rates are 0.05 (prefix), 0.01 (middle), 0.39 (suffix). Only suffix placement elevates follow-wrong (suffix-prefix: $p < 0.001$). To distinguish answer-text *content* from end-of-chain *position/recency*, we ran a neutral-filler control ($N=100$ suffix examples from the same counterbalanced set): the conflicting suffix step was replaced by a content-free sentence (“The computation above confirms the reasoning.”) that occupies the identical suffix position without encoding any answer. In a matched $N=100$ subsample, the conflicting-suffix condition replicated follow-wrong at 0.32 (consistent with the main experiment 0.39, $N=500$); the neutral-filler condition dropped to 0.00 (0 out of 100 examples), confirming that the suffix override is driven by answer-text *content*, not end-of-chain position or recency.

A.5 Self-Generated Chain Experiment

All preceding experiments operate in the *consumption* setting: the model receives a pre-existing chain and produces a final answer. A natural objection is that models may behave differently when consuming chains they did not generate. To address this, we test whether the rationalization signature persists when the model evaluates *its own* generated chains of thought.

Design. We run Qwen 2.5-3B-Instruct on 300 GSM8K examples in a two-phase protocol:

1. *Generation phase.* The model generates a complete chain of thought for each problem. We retain only examples where the self-generated chain produces the correct answer ($N_{\text{correct}} = 147$; generation accuracy = 0.49).
2. *Consumption phase.* For each correctly-solved example, we take the model’s *own* generated steps and apply the same three-condition protocol from Section 7.2:
 - SG-SC: self-generated steps + correct answer line,
 - SG-CC: self-generated steps + conflicting wrong answer,
 - QO: question only (no chain).

If the consumption objection holds, i.e., the model reasons more faithfully over its own chains, the followed-wrong rate under SG-CC should be substantially lower than the pre-written CC rate of 0.63. We test this directly with $N=300$ examples.

Results. Table 11 summarizes the self-generated chain results.

Table 11: Self-generated chain experiment on Qwen 2.5-3B-Instruct. The model first generates its own CoT (generation accuracy 0.49, $N_{\text{correct}}=147$), then consumes its own steps under the standard three-condition protocol. ACC: fraction correct. FW: fraction following the wrong explicit answer (SG-CC only).

Condition	Acc	Followed-wrong (FW)
Self-gen standard (SG-SC)	0.88	—
Self-gen conflicting (SG-CC)	0.19	0.69
Question only (QO)	0.12	—

Even when the model evaluates chains it has *just generated itself*, the followed-wrong rate under SG-CC is 0.69, statistically indistinguishable from the pre-written rate (0.63) and far above zero ($p < 10^{-5}$ Fisher exact vs. question-only; Wilson 95% CI [0.61, 0.76]). Critically, 0.69 is statistically indistinguishable from the pre-written chain rate (0.63; Fisher exact $p=0.240$, two-sided), confirming that the consumption-time override is not an artifact of chain provenance. The answer-text tracking behavior persists when the model evaluates its own reasoning, ruling out the consumption objection.

A.6 Commonsense Reasoning Replication

To test whether the conflicting-answer answer-text override effect generalizes beyond arithmetic, we apply the same three-condition protocol to 150 commonsense reasoning examples spanning five domains: social reasoning, temporal reasoning, counterfactual reasoning, spatial reasoning, and

causal reasoning. Unlike the GSM8K items, these examples have *text-based* answers (e.g., “grateful,” “the fire grows”), eliminating the possibility that the model merely pattern-matches digits from the chain.

Results. Table 12 reports the cross-domain results. Under the standard chain (SC), the model achieves accuracy 0.82, confirming that the chain aids reasoning on commonsense tasks. Under the conflicting chain (CC), accuracy drops to 0.10, with a followed-wrong rate of 0.76, the model follows the injected wrong answer in 76% of trials (one-sided binomial test versus $H_0: fw=0.5$: $p<10^{-10}$). The question-only baseline confirms that the model cannot solve these items without reasoning context (QO accuracy = 0.09).

Notably, the commonsense followed-wrong rate (0.76) is *higher* than the arithmetic rate (0.63), suggesting that the answer-tracking mechanism is at least as strong for natural-language answers as for numeric ones. This rules out the hypothesis that the conflicting-answer effect is specific to answer-format extraction (e.g., the model merely copying a trailing digit).

Table 12: Conflicting-answer experiment on commonsense reasoning (Qwen 2.5-3B-Instruct, $N=150$, 5 domains). Text-based answers confirm the answer-text override effect generalizes beyond arithmetic.

Condition	Accuracy	Followed-wrong (FW)
Standard chain (SC)	0.82	—
Conflicting chain (CC)	0.10	0.76
Question only (QO)	0.09	—

Scale comparison: 7B commonsense. Extending the commonsense conflicting-answer test to Qwen 2.5-7B-Instruct (same 150 examples, same protocol) yields a followed-wrong rate of 0.660, compared to 0.76 at 3B scale. Standard-chain accuracy is 0.853 (vs. 0.82 at 3B), confirming the 7B model reasons well on commonsense items. The answer-text override effect is substantial at both scales, demonstrating that answer-text override on commonsense reasoning is robust across model size and consistent with the override observed on arithmetic tasks. Question-only accuracy remains low (0.133 vs. 0.09 at 3B), confirming that the 7B model’s high chain accuracy reflects genuine reasoning from steps rather than question-only competence.

Commonsense format ablation. To test whether the format-determination pattern replicates on non-arithmetic data, we apply a five-condition positional ablation to the same 150 commonsense examples (Table 13). Each chain’s final step contains the explicit answer; the penultimate step contains supporting reasoning. Corrupting the answer-bearing final step collapses accuracy from 0.82 to 0.29 (−54.0 pp), while corrupting the penultimate reasoning step has no measurable effect (0.83 vs. 0.82; $p=1.0$, paired sign test). The penultimate–last gap is 54.0 pp ($p<10^{-22}$), confirming that positional sensitivity tracks answer placement on text-answer commonsense data, not just arithmetic.

Table 13: Commonsense format ablation (Qwen 2.5-3B-Instruct, $N=150$). Corrupting the answer-bearing final step collapses accuracy; corrupting the reasoning penultimate step does not.

Condition	Accuracy	Δ vs. full chain
Full chain	0.82	—
Corrupt penultimate (reasoning)	0.83	−0.01
Neutral-stripped (last removed)	0.32	−0.50
No conclusion (last deleted)	0.35	−0.47
Corrupt last (answer step)	0.29	−0.53

A.7 Distillation Analysis (Exploratory)

Distillation hypothesis (preliminary). DeepSeek-R1-Distill-Qwen-7B³ (Qwen-2.5-Math-7B fine-tuned on DeepSeek-R1’s reasoning traces; $N=100$) achieves 0.990 standard-chain accuracy and

³HuggingFace path: `deepseek-ai/DeepSeek-R1-Distill-Qwen-7B`.

0.010 question-only accuracy. The followed-wrong rate is 0.980 ($p < 10^{-12}$), matching the Qwen-7B base rate (FW = 1.00; protocol-matched v1 comparison: FW = 0.590) and far exceeding the Phi-4-14B rate (FW = 0.300). This single-pair comparison is consistent with a preliminary hypothesis that parameter scale, rather than reasoning-trace supervision, is the primary determinant of override resistance, but the comparison conflates scale with training distribution and should be treated as exploratory rather than confirmatory.

A.8 Base Model Comparison

Base model comparison (instruction-following confound test). To directly test whether the override effect depends on instruction-tuning (RLHF/supervised fine-tuning), we evaluated Qwen 2.5-7B, the non-instruction-tuned base model with identical architecture and pretraining, on the GSM8K-v1 protocol ($N=100$, same examples as EXP-28A). SC accuracy 0.990, CC accuracy 0.090, followed-wrong rate 0.900 ($p < 10^{-10}$), question-only accuracy 0.000. The base model’s followed-wrong rate (FW = 0.900) exceeds that of the instruction-tuned counterpart (FW = 0.590), indicating that instruction tuning partially mitigates rather than creates the override effect. The override is a property of the pretrained representations: Qwen-7B-Base, which has not been fine-tuned on instruction-following tasks, shows an even stronger tendency to defer to an explicit conflicting answer statement than the RLHF-tuned model. This rules out RLHF as the primary driver and places the phenomenon in the pretraining regime. The modest reduction from base to instruct ($\Delta = 0.310$) suggests instruction tuning provides limited resistance to answer-text override.